Voluntary Feature Revelation in Predictive Tasks: A Unified Framework for Regression and Multiclass Classification

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We investigate the framework of machine learning in the presence of strategic information disclosure, specifically focusing on the problem of voluntary feature revelation. This scenario is modeled as a Stackelberg game: a data provider (the sender) strategically determines whether to reveal or withhold optional features to optimize their individual payoff, while a decision-maker (the receiver) commits to a predictive model designed to minimize strategic prediction error. We analyze this interaction across two fundamental prediction spaces: continuous labels (regression) and discrete labels (multiclass classification). For both settings, we examine three sequential regimes of interaction: (i) the decision-theoretic game foundations, establishing necessary and sufficient conditions under which voluntary disclosure benefits the receiver; (ii) the offline batch learning problem, deriving uniform generalization bounds and sample complexity rates under finite training data; and (iii) the online learning problem, designing algorithms to learn under partial or bandit feedback in real-time. Since optimizing the discontinuous, step-like strategic risk is computationally intractable (NP-hard), we leverage smooth surrogate loss formulations to construct gradient-based optimization routines with rigorous regret guarantees—proving $O(\sqrt{T})$ regret for regression and $O(T^{2/3})$ expected regret for classification. Our theoretical findings are validated through comprehensive simulations, mapping out strategic advantage regions and confirming the empirical convergence rates under varying environment dynamics.

Additional Key Words and Phrases: Strategic Machine Learning, Voluntary Disclosure, Stackelberg Games, Sample Complexity, Online Regret, Surrogate Loss

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1 Introduction

Traditional machine learning paradigms operate under the fundamental assumption that data features are passively observed and completely reported. In many socio-economic applications, however, the data is generated by strategic agents who possess private preferences and can actively manipulate the information they share. When a decision-maker commits to a predictive model, these agents strategically decide which features to reveal and which to withhold to steer the model's prediction in their favor. This phenomenon of voluntary disclosure under strategic feature

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withholding is common in markets characterized by asymmetric information, where the decision-maker must predict an outcome based on partially self-reported feature vectors.

To illustrate the multi-directional incentives of such strategic withholding, consider an individual navigating the insurance sector. When purchasing a health or life insurance policy, the applicant seeks to minimize their premium. They have a strategic incentive to withhold negative health indicators—such as chronic conditions or high-risk habits—to present an understated risk profile to the underwriter. Conversely, if the same individual later files a claim for disability benefits or worker's compensation, their preference flips: they now seek to maximize the payout, motivating them to selectively disclose their medical history to present an overstated severity of their condition. In both scenarios, the observed features are not missing at random; they are the output of a utility-maximizing disclosure strategy that directly responds to the committed prediction rule.

Crucially, strategic feature withholding is not universally detrimental or beneficial to the system. Whether voluntary disclosure improves or degrades the decision-maker's prediction accuracy depends heavily on the alignment between the sender's preferences and the true target, the inherent value of the features, and external channel noise. We are interested in formally studying and quantifying these dynamics. This motivates a dual-level investigation: first, a *decision-theoretic* game analysis to characterize when voluntary disclosure yields a positive strategic advantage over mandatory disclosure; and second, a *learning-theoretic* analysis to determine how the decision-maker can learn the optimal predictive rules under finite sample regimes (offline) and adapt dynamically in real-time under partial or bandit feedback (online).

In this paper, we establish a comprehensive set of theoretical results across the six settings of strategic feature withholding, outlining the fundamental limits, sample complexities, and online regret behaviors. For the regression setting, we establish necessary and sufficient conditions for the decision-theoretic game under the structural assumption of an all-or-nothing disclosure setup where agents reveal the entire feature vector or withhold it completely. We show that when the sender's preferences are aligned with the true target, the strategic advantage is positive because the sender naturally reveals features when they are most predictive. In the offline batch learning regime, we prove that the strategic empirical risk minimizer (ERM) on a smooth surrogate loss is consistent and achieves a uniform generalization rate of $O(d^3/\sqrt{n})$ under finite training samples. For the online setting, we design a gradient-free online learning algorithm that operates under bandit feedback and achieves a cumulative regret of $O(d\sqrt{T})$, establishing that the receiver can dynamically learn the strategic prediction rule in real-time despite the partial observability of features.

For the multiclass classification setting, we show that the strategic risk is an affine function of the channel noise, where the optimal default class partitions the noise space into a piecewise-linear envelope of at most K subintervals. Under the same all-or-nothing disclosure assumption, we formulate a joint offline learning algorithm and prove that the excess risk decays at a rate of $O(\sqrt{Kd\log n/n})$ based on the Natarajan dimension of the hypothesis space. In the online sequential classification game, we propose a decoupled EXP3+OGD algorithm that explores the discrete default classes in a bandit fashion while optimizing the continuous predictors, proving an expected cumulative regret of $\tilde{O}(T^{2/3})$ under a non-degenerate mass condition on the revealed subpopulation.

1.1 Related Work

Our work lies at the confluence of microeconomic information design, strategic classification, and learning theory. Rather than analyzing these topics in isolation, we position our work by tracing how our unified framework builds upon and extends three major thematic areas in the literature.

First, our work is related to the literature on voluntary disclosure and information design. The economic foundations of voluntary disclosure under verifiable information were pioneered by Milgrom [?] and Grossman [?], who established the classical "unraveling" result: when information can be verified, a rational receiver assumes the worst about any withheld information, forcing the sender to disclose all private details in equilibrium. However, complete unraveling fails under realistic market frictions. Dye [?] and Verrecchia [?] showed that if there is uncertainty regarding whether the sender possesses the information (often modeled as an erasure channel) or if disclosure incurs proprietary costs, senders will strategically withhold unfavorable information. These models were extended to asset pricing by Jovanovic [?] and Shin [?], and generalized under the framework of Bayesian persuasion by Kamenica and Gentzkow [?]. While this literature focuses on low-dimensional signals with fixed sender-receiver actions, our work extends these voluntary disclosure models to high-dimensional feature spaces ($X \in \mathbb{R}^d$) and analyzes how a predictive model can be learned. We establish the necessary and sufficient conditions under which voluntary feature disclosure provides a strategic advantage to the receiver, extending Dye's classical results to multidimensional regression and multiclass classification settings.

Second, our work connects to strategic classification and performative prediction. Within computer science, the interaction between strategic agents and predictive models is studied under strategic classification, pioneered by Hardt et al. [?]. In their Stackelberg formulation, agents pay a cost to continuously alter their features to obtain a favorable classification. This has been extended to study algorithmic recourse by Tsirtsis and Rodriguez [?], causal strategic models by Shavit et al. [?], and performative prediction by Perdomo et al. [?] and Miller et al. [?], which analyze how predictions shift the underlying distribution. Further, researchers have investigated the social costs [?] and fairness implications [?] of such models. In contrast to standard strategic classification—where agents continuously perturb features at a cost—our framework focuses on a discrete action space where agents strategically withhold or reveal their entire verifiable feature vector. This discrete choice introduces non-convexities and step-like strategic risks. We show how these discrete interactions partition the noise space $[0, 1]$ into piecewise-linear optimal default regions and analyze how the receiver can learn under these discontinuous risk landscapes.

Third, our work is positioned within the literature on learning and optimization under strategic constraints. Dekel et al. [?] and Chen et al. [?] studied offline incentive-compatible regression where agents strategically manipulate their target labels or features. In online settings, Chen et al. [?] and Dong et al. [?] analyzed online gradient updates when learning from strategic agents. Under bandit feedback or partial observability, online convex optimization relies on gradient-free techniques, such as the one-point bandit gradient estimator of Flaxman et al. [?], and multi-armed bandit dynamics as reviewed in Auer et al. [?] and Cesa-Bianchi and Lugosi [?]. We bridge these domains by deriving the first uniform generalization bounds for strategic feature withholding in both offline regression (via product-class Rademacher complexity) and multiclass classification (via Natarajan dimension). In online settings under bandit feedback—where features are hidden when agents withhold—we design updates that combine Flaxman's bandit gradient descent for regression and a decoupled EXP3+OGD algorithm for classification, proving sublinear cumulative regret bounds.

2 Problem Formulation

We model the voluntary feature disclosure setting as a sequential Stackelberg game between two players: a receiver (the decision-maker) who acts as the leader, and a population of senders (data providers) who act as the followers. Let $\mathcal{P}$ denote a joint probability distribution over the triplet (U, X, Y) representing a random sender from the population, where $X \in \mathcal{X} \subseteq \mathbb{R}^d$ is the sender's private, high-dimensional feature vector, $Y \in \mathcal{Y}$ is the true ground-truth label that the receiver aims to predict, and $U \in \mathcal{U}$ represents the sender's private target value or preference for the receiver's prediction. The prediction space is denoted by $\mathcal{Y}$, which we specialize to the continuous space $\mathcal{Y} = \mathbb{R}$ for regression in Section 3 and the discrete multi-class space $\mathcal{Y} = \{1, \ldots, K\}$ for classification in Section 4.

The game unfolds sequentially and is characterized by the following mathematical operations. The receiver first commits to a predictive policy $H = (h, d)$, wherein $h \in \mathcal{H} \subset \{f : \mathcal{X} \to \mathcal{Y}\}$ acts as the feature-based predictor and $d \in \mathcal{Y}$ serves as the default prediction for withheld features. A sender, with private characteristics $(U, X, Y) \sim \mathcal{P}$, then observes H and determines their disclosure strategy mapping $D : \mathcal{U} \times \mathcal{X} \to \{0, 1\}$. To capture channel imperfections, the transmission passes through a memoryless erasure channel denoted by a known erasure probability $q \in [0, 1]$. The final revelation outcome is governed by the random variable $A = D(U, X) \cdot (1 - \epsilon)$, where $\epsilon \sim \text{Bernoulli}(q)$ represents an independent erasure event ($\epsilon = 1$ if erased, $\epsilon = 0$ if successfully transmitted). The receiver subsequently observes the incoming message $M \in \mathcal{X} \cup \{\emptyset\}$, yielding the prediction $\hat{Y} = H(M) = Ah(X) + (1 - A)d$.

Both players behave rationally to minimize their respective expected losses. A utility-maximizing sender selects their disclosure decision to minimize their expected loss over the channel erasure, defined as $D^*(H; U, X) \in \arg\min_{D \in \{0,1\}} \mathbb{E}_\epsilon \left[\ell_S \left(D(1-\epsilon)h(X) + (1 - D(1-\epsilon))d,\ U\right)\right]$, breaking ties in favor of withholding ($D = 0$). Senders evaluate the prediction $\hat{Y}$ relative to their private preference U using a loss function $\ell_S : \mathcal{Y} \times \mathcal{U} \to \mathbb{R}_+$. Conversely, the receiver evaluates the prediction relative to the true label Y using a loss function $\ell_R : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+$. The senders' best responses induce a joint distribution on the observed message M and label Y. The receiver's objective is to minimize their strategic risk $R_S(h, d; q)$, defined as the expected receiver loss under the sender's best-response strategy:

$$R_S(h, d; q) = \mathbb{E}_{(U,X,Y)\sim\mathcal{P}} \left[\mathbb{E}_\epsilon \left[\ell_R \left(Ah(X) + (1 - A)d,\ Y\right)\right]\right], \tag{1}$$

where the revelation outcome is determined by $A = D^*(H; U, X)\cdot(1-\epsilon)$. The Stackelberg equilibrium of this game is defined by the receiver's optimal policy (h^*, d^*) that minimizes this strategic risk:

$$(h^*, d^*) \in \underset{h\in\mathcal{H},\, d\in\mathcal{Y}}{\arg\min}\ R_S(h, d; q), \tag{2}$$

with the optimal strategic risk denoted by $R_S^*(q) = R_S(h^*, d^*; q)$. The environment of the game is fully characterized by the tuple $E = (\mathcal{P}, \mathcal{H}, \mathcal{Y}, q, \ell_S, \ell_R)$.

3 Regression under Strategic Withholding

In this section, we specialize the general voluntary disclosure model to continuous prediction spaces where the labels are real-valued, $\mathcal{Y} = \mathbb{R}$, and features lie in a vector space $\mathcal{X} \subseteq \mathbb{R}^d$. We focus on linear predictors under quadratic loss, reflecting the classical setting of strategic linear regression. We analyze three sequential regimes of interaction: first, the decision-theoretic game foundations under a known joint distribution, characterizing when voluntary feature disclosure provides a strategic advantage over a baseline of mandatory disclosure; second, the offline batch learning problem under an unknown distribution, analyzing how many training samples are needed to learn the optimal model; and third, the online sequential learning problem under bandit feedback, designing algorithms to update the receiver's model dynamically in real-time.

To analyze the decision-theoretic game, we contrast the voluntary disclosure model with a baseline of mandatory disclosure, where the sender is contractually obligated to always reveal their features. Under this baseline, the feature vector X is revealed unless it is lost in transmission. For any receiver strategy $H_b = (b, d)$ with a linear predictor $h_b(x) = b^\top x$ and default prediction $d = 0$, the mandatory disclosure risk (vanilla risk) is $R_V(b,q) = (1-q)\mathbb{E}_{(X,Y)\sim\mathcal{P}}[(b^\top X - Y)^2] + q\mathbb{E}_{Y\sim\mathcal{P}}[Y^2]$. The optimal mandatory risk is $R_V^*(q) = \min_{b\in\mathbb{R}^d} R_V(b,q)$. We define the strategic advantage $\Delta(E)$ of the environment as the difference in prediction risk achieved by voluntary disclosure relative to mandatory disclosure: $\Delta(E) = R_V^*(q) - R_S^*(q)$, where $R_S^*(q)$ is the optimal strategic risk. A positive strategic advantage ($\Delta(E) > 0$) implies that voluntary disclosure is strictly beneficial to the receiver.

Under learning settings where the joint distribution $\mathcal{P}$ is unknown, the receiver must estimate the optimal predictor from data. In the offline batch learning setup, the receiver has access to a training set of n i.i.d. samples and seeks to find an empirical predictor that minimizes the true strategic risk, bounding the generalization gap $R_S(\hat{b},q) - R_S(b^*,q)$ as a function of the sample size n. In the online sequential learning setup under partial or bandit feedback, the receiver updates the parameter b_t over rounds $t = 1,\ldots,T$ based only on the observed message $M_t \in \{x_t, \emptyset\}$ and the true label y_t (without observing the sender's private preference u_t), aiming to minimize the expected cumulative regret $\mathrm{Regret}(T) = \mathbb{E}[\sum_{t=1}^T R_S(b_t,q)] - \min_b \sum_{t=1}^T R_S(b,q)$.

3.1 The Regression Decision Problem

Let $(U,X,Y) \sim \mathcal{P}$ be a joint random variable representing the sender's preference, features, and true labels, respectively. We assume that all variables are zero-mean, i.e., $\mathbb{E}[X] = 0$, $\mathbb{E}[Y] = 0$, and $\mathbb{E}[U] = 0$.

The receiver committed strategy space consists of linear predictors that predict 0 on empty messages and $b^\top x$ on revealed messages:

$$\mathcal{H} = \{h_b : h_b(\emptyset) = 0,\ h_b(x) = b^\top x,\ b \in \mathbb{R}^d\}. \tag{3}$$

Both players evaluate outcomes using the squared loss function:

$$\ell_R(\hat{y},y) = (\hat{y}-y)^2, \quad \ell_S(\hat{y},u) = (\hat{y}-u)^2. \tag{4}$$

For a given receiver slope parameter $b \in \mathbb{R}^d$, we define two central statistics that dictate the game dynamics:

$$F_b(U,X) = (2U - b^\top X)b^\top X, \tag{5}$$

$$G_b(X,Y) = (2Y - b^\top X)b^\top X. \tag{6}$$

The statistic $F_b(U,X)$ governs the sender's disclosure decision, while $G_b(X,Y)$ measures the receiver's error reduction when features are successfully disclosed, representing $Y^2 - (Y - b^\top X)^2$.

We first characterize the baseline mandatory disclosure risk.

Lemma 3.1 (Vanilla Risk Decomposition). *For any receiver slope $b \in \mathbb{R}^d$ and erasure probability q, the vanilla risk under mandatory disclosure decomposes as:*

$$R_{\mathrm{V}}(b,q) = \mathrm{Var}(Y) - (1-q)\mathrm{Var}(b_{\mathrm{V}}^\top X) + (1-q)\mathrm{Var}((b-b_{\mathrm{V}})^\top X), \tag{7}$$

where $b_{\mathrm{V}} = \mathrm{Var}(X)^{-1}\mathrm{Cov}(X,Y)$ is the ordinary least-squares (OLS) slope. The optimal vanilla risk is:

$$R_{\mathrm{V}}^*(q) = \mathrm{Var}(Y) - (1-q)\mathrm{Var}(b_{\mathrm{V}}^\top X). \tag{8}$$

The proof is postponed to Section 7.1.

Next, we characterize the strategic best response of the sender follower.

Lemma 3.2 (Sender Best Response). *For a committed receiver strategy $b \in \mathbb{R}^d$, the sender's optimal disclosure decision is:*

$$D^*(b;U,X) = \mathbf{1}\{F_b(U,X) > 0\}. \tag{9}$$

That is, the sender reveals features if and only if $F_b(U,X) > 0$.

The proof is postponed to Section 7.1.

Using the sender's best response, the receiver's strategic risk can be formulated as follows.

Lemma 3.3 (Strategic Risk Formula). *For any slope parameter $b \in \mathbb{R}^d$ and channel erasure probability q, the strategic risk is:*

$$R_{\mathrm{S}}(b,q) = \mathrm{Var}(Y) - (1-q)\mathbb{E}\left[G_b(X,Y)\mathbf{1}\{F_b(U,X) > 0\}\right]. \tag{10}$$

The proof is postponed to Section 7.1.

By comparing the optimal vanilla risk $R^*_{\mathrm{V}}(q)$ and the strategic risk $R_{\mathrm{S}}(b,q)$, we derive the conditions under which voluntary disclosure is strictly beneficial to the receiver, yielding a positive strategic advantage:

$$\Delta(E) = R^*_{\mathrm{V}}(q) - \min_{b\in\mathbb{R}^d} R_{\mathrm{S}}(b,q) > 0. \tag{11}$$

We establish a sufficient condition for voluntary disclosure to outperform mandatory disclosure.

Theorem 3.4 (Sufficient Condition for Strategic Advantage). *For a given environment E, the strategic advantage is positive if there exists a vector $b \in \mathbb{R}^d$ such that:*

$$\|U - Y\|_2 < \frac{\mathbb{E}[|G_b(X,Y)|] - \mathrm{Var}(b_{\mathrm{V}}^\top X) - \mathrm{Var}((b-b_{\mathrm{V}})^\top X)}{4\sqrt{\mathrm{Var}(b^\top X)}}, \tag{12}$$

where $\|U - Y\|_2 = \sqrt{\mathbb{E}[(U-Y)^2]}$ represents the alignment of the preference value U with the ground truth Y.

The proof is postponed to Section 7.1.

We also establish a matching necessary condition.

Theorem 3.5 (Necessary Condition for Strategic Advantage). *A necessary condition for the strategic advantage to be positive is that there exists a vector $b \in \mathbb{R}^d$ satisfying:*

$$\mathbb{E}[|G_b(X,Y)|] - \mathrm{Var}(b_{\mathrm{V}}^\top X) - \mathrm{Var}((b-b_{\mathrm{V}})^\top X) > 0. \tag{13}$$

The proof is postponed to Section 7.1.

Remark 1. *Crucially, the channel noise parameter q acts only as a scaling factor on the strategic advantage and does not affect the validity of the sufficient or necessary conditions. If Theorem 3.5 fails, voluntary disclosure cannot yield lower risk than mandatory disclosure for any noise level $q \in (0,1)$.*

3.2 Offline Learning (Batch Sample Complexity)

When the underlying joint distribution $\mathcal{P}$ is unknown, the receiver must estimate the optimal predictor from a training set of n i.i.d. samples $S = \{(u_i, x_i, y_i)\}_{i=1}^n$.

To prevent overfitting and establish uniform bounds, we assume that the data support is bounded, i.e., $\|x_i\|_\infty \le C$, $|u_i| \le C$, and $|y_i| \le C$ almost surely for a constant $C > 0$. The receiver restricts slope parameters to a bounded domain:

$$\mathcal{W} = \{b \in \mathbb{R}^d : \|b\|_\infty \le C\}. \tag{14}$$

For any committed slope b, we define the empirical vanilla risk and empirical strategic risk over the sample S as:

$$\hat{R}_{\mathrm{V}}(b,q) = \frac{1}{n}\sum_{i=1}^n y_i^2 - (1-q)\frac{1}{n}\sum_{i=1}^n (y_i - b^\top x_i)^2, \tag{15}$$

$$\hat{R}_{\mathrm{S}}(b,q) = \frac{1}{n}\sum_{i=1}^n y_i^2 - (1-q)\frac{1}{n}\sum_{i=1}^n G_b(x_i,y_i)\mathbf{1}\{F_b(u_i,x_i) > 0\}. \tag{16}$$

The empirical optimal estimators are defined by:

$$\hat{b}_{\mathrm{V}} \in \arg\min_{b \in \mathcal{W}} \hat{R}_{\mathrm{V}}(b, q), \quad \hat{b}_{\mathrm{S}} \in \arg\min_{b \in \mathcal{W}} \hat{R}_{\mathrm{S}}(b, q). \tag{17}$$

The receiver executes the offline empirical risk minimization algorithm detailed in Algorithm 1.

Algorithm 1 Strategic Regression Empirical Risk Minimization (SR-ERM)

1: **Input:** Training set $S = \{(u_i, x_i, y_i)\}_{i=1}^n$, channel erasure probability q, bounded domain parameter C.

2: **Define:** Strategy space $\mathcal{W} = \{b \in \mathbb{R}^d : \|b\|_\infty \leq C\}$.

3: **Define:** Functions $F_b(u_i, x_i) = (2u_i - b^\top x_i)b^\top x_i$ and $G_b(x_i, y_i) = (2y_i - b^\top x_i)b^\top x_i$ for all $i \in \{1, \ldots, n\}$.

4: **Formulate Empirical Risks:**

$$\hat{R}_{\mathrm{V}}(b, q) = \frac{1}{n}\sum_{i=1}^n y_i^2 - (1-q)\frac{1}{n}\sum_{i=1}^n (y_i - b^\top x_i)^2$$

$$\hat{R}_{\mathrm{S}}(b, q) = \frac{1}{n}\sum_{i=1}^n y_i^2 - (1-q)\frac{1}{n}\sum_{i=1}^n G_b(x_i, y_i)\mathbf{1}\{F_b(u_i, x_i) > 0\}$$

5: **Compute Empirical Strategic Advantage:**

$$\hat{\Delta}(S) = \min_{b \in \mathcal{W}} \hat{R}_{\mathrm{V}}(b, q) - \min_{b \in \mathcal{W}} \hat{R}_{\mathrm{S}}(b, q)$$

6: **Output:** Optimal empirical predictors $\hat{b}_{\mathrm{V}}$ and $\hat{b}_{\mathrm{S}}$, and strategic advantage estimate $\hat{\Delta}(S)$.

We derive uniform generalization bounds for both disclosure regimes.

Lemma 3.6 (Uniform Generalization of Vanilla Risk). *Let $\delta \in (0, 1)$. With probability at least $1 - \delta$ over the random draw of S:*

$$\sup_{b \in \mathcal{W}} |R_{\mathrm{V}}(b, q) - \hat{R}_{\mathrm{V}}(b, q)| \leq O\left(\frac{d^2 C^4}{\sqrt{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\right). \tag{18}$$

The proof is postponed to Section 7.1.

Lemma 3.7 (Uniform Generalization of Strategic Risk). *Let $\delta \in (0, 1)$. With probability at least $1 - \delta$ over the random draw of S:*

$$\sup_{b \in \mathcal{W}} |R_{\mathrm{S}}(b, q) - \hat{R}_{\mathrm{S}}(b, q)| \leq O\left(d^3 C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\right). \tag{19}$$

The proof is postponed to Section 7.1.

These bounds enable us to show that the empirical strategic advantage $\hat{\Delta}(S) = \min_b \hat{R}_{\mathrm{V}}(b, q) - \min_b \hat{R}_{\mathrm{S}}(b, q)$ is a consistent estimator of the true strategic advantage.

Theorem 3.8 (Generalization Bound for Strategic Advantage). *Let $\delta \in (0, 1)$. With probability at least $1 - \delta$ over the random draw of S:*

$$|\Delta(E) - \hat{\Delta}(S)| \leq O\left(d^3 C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\right). \tag{20}$$

The proof is postponed to Section 7.1.

Combining these generalization limits with a stability analysis of the decision boundary, we bound the excess risk of the joint learning algorithm.

THEOREM 3.9 (EXCESS RISK OF THE JOINT LEARNER). *Let $\delta \in (0, 1)$. The excess strategic risk of the joint selection algorithm $\mathcal{A}$ relative to the optimal oracle choice satisfies:*

$$\mathbb{E}_S[R(\mathcal{A}, S)] \leq O\left(d^3 C^4 \sqrt{\frac{\log n}{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right). \tag{21}$$

The proof is postponed to Section 7.1.

3.3 Online Learning (Sequential Adaptive Control)

We now study the online learning problem where the receiver sequentially updates their linear predictor weight $b_t \in \mathcal{W}$ over rounds $t = 1, \ldots, T$. To separate the channel friction from the learning dynamics, we assume $q = 0$ in this subsection (i.e., a noiseless channel where voluntary disclosure determines revelation directly).

At each round t, the receiver commits to weight $b_t \in \mathcal{W}$ and nature draws the private sender characteristics $(u_t, x_t, y_t) \sim \mathcal{P}$. Senders disclose their features with a probability governed by a smooth, ρ-Lipschitz approximation $\psi : \mathbb{R} \to [0, 1]$ of the step function, so that the revelation indicator is drawn as $A_t \sim \text{Bernoulli}(\psi(F_{b_t}(u_t, x_t)))$ with $F_{b_t}(u_t, x_t) = (2u_t - b_t^\top x_t) b_t^\top x_t$. The receiver then observes the incoming message $M_t = x_t$ if successfully revealed ($A_t = 1$), and $M_t = \emptyset$ if withheld ($A_t = 0$), outputting the prediction $\hat{y}_t = A_t b_t^\top x_t$ and suffering the quadratic loss $\ell_t(b_t) = (1 - A_t) y_t^2 + A_t (y_t - b_t^\top x_t)^2$.

The true strategic population risk is:

$$L(b) = \mathbb{E}_{(U,X,Y)\sim\mathcal{P}}\left[Y^2 - \psi(F_b(U, X)) G_b(X, Y)\right], \tag{22}$$

where $G_b(X, Y)$ is defined as in (6). The gradient of this non-convex risk is derived below.

LEMMA 3.10 (POPULATION GRADIENT FORMULA). *The gradient of the strategic population risk is:*

$$\nabla L(b) = -2\mathbb{E}\left[\psi'(F_b(U, X))(U - b^\top X) G_b(X, Y) X + \psi(F_b(U, X))(Y - b^\top X) X\right], \tag{23}$$

where $\psi'(z) = \frac{d\psi(z)}{dz}$.

The proof is postponed to Section 7.1.

We analyze learning guarantees under a bandit feedback model. To eliminate the asymptotic bias caused by hidden sender preferences U without observing them directly, we can optimize $L(b)$ directly using a one-point bandit gradient estimator. While we do not observe U or unrevealed features, the scalar loss $c_t = \ell_t(b_t)$ is always observable: $c_t = y_t^2$ if $A_t = 0$, and $c_t = (y_t - b_t^\top x_t)^2$ if $A_t = 1$.

At each round t, the receiver perturbs the slope parameter:

$$\hat{b}_t = b_t + \delta_t v_t, \quad v_t \sim \text{Uniform}(\mathbb{S}^{d-1}), \tag{24}$$

where $\delta_t > 0$ is the exploration radius. The receiver observes $c_t = \ell_t(\hat{b}_t)$ and updates the center parameter b_t using the bandit gradient estimator: The sequential update procedure under bandit feedback is detailed in Algorithm 2.

THEOREM 3.11 (BANDIT-FEEDBACK REGRET BOUND). *Assume that the true risk $L(b)$ is locally λ-strongly convex and H-smooth. By running the bandit gradient descent algorithm with learning rate*

Algorithm 2 Strategic Bandit Online Gradient Descent (SB-OGD)

Initialize: Parameter domain $\mathcal{W} = \{b \in \mathbb{R}^d : \|b\|_2 \leq R\}$, center point $b_1 \in \mathcal{W}$, step-size parameters η_t, exploration radii δ_t.
for $t = 1, \dots, T$ **do**
  Sample direction vector $v_t \sim \text{Uniform}(\mathbb{S}^{d-1})$ on the unit sphere.
  Formulate perturbed parameter: $\hat{b}_t = b_t + \delta_t v_t$.
  Commit to policy parameterized by $\hat{b}_t$ and observe sender's revelation outcome $A_t \in \{0, 1\}$.
  Suffer and observe scalar loss:

$$c_t = \begin{cases} y_t^2 & \text{if } A_t = 0, \\ (y_t - \hat{b}_t^\top x_t)^2 & \text{if } A_t = 1. \end{cases}$$

  Compute unbiased bandit gradient estimate: $\hat{g}_t = \frac{d}{\delta_t} c_t v_t$.
  Update center parameter: $b_{t+1} = \Pi_{\mathcal{W}}(b_t - \eta_t \hat{g}_t)$.
end for

$\eta_t = \frac{1}{\lambda t}$ *and exploration radius* $\delta_t = \delta_0 t^{-1/4}$*, the expected average strategic regret satisfies:*

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[L(\hat{b}_t) - L(b^*)] \leq O\left(\frac{d}{\sqrt{T}}\right). \tag{25}$$

The proof is postponed to Section 7.1.

Remark 2. *The bandit feedback algorithm eliminates the landscape bias floor ϵ entirely at the cost of a dimension-dependent scaling factor d and variance inflation. This establishes a clean rate-bias trade-off for online learning under strategic feature withholding.*

4 Multiclass Classification under Strategic Withholding

In this section, we specialize the general voluntary disclosure model to discrete prediction spaces where the outcome labels are finite, $\mathcal{Y} = \{1, \dots, K\}$, and the sender's preferences are also discrete, $\mathcal{U} = \{1, \dots, K\}$. We analyze this interaction under the standard 0-1 loss function, which poses unique mathematical challenges due to its discontinuity. We analyze three sequential regimes of interaction: first, the decision-theoretic game foundations under a known joint distribution, showing how the optimal default class partitions the unit interval of channel noise; second, the offline batch learning problem under an unknown distribution, deriving the sample complexity for a joint default-predictor learner; and third, the online sequential learning problem, presenting an adaptive algorithm to update both the default class and the predictor weights in real-time under bandit feedback.

To analyze the decision-theoretic classification game, we contrast the strategic setting with a baseline of mandatory disclosure, where the sender is contractually obligated to always reveal their features. For any default class $i \in \{1, \dots, K\}$ and classifier $h : \mathcal{X} \to \{1, \dots, K\}$, the mandatory disclosure risk (vanilla risk) is $R_V(h, i; q) = (1 - q)\mathbb{E}_{(X,Y)\sim\mathcal{P}}[\mathbf{1}\{h(X) \neq Y\}] + q\mathbb{E}_{Y\sim\mathcal{P}}[\mathbf{1}\{i \neq Y\}]$. The optimal vanilla risk is $R_V^*(q) = \min_{h,i} R_V(h, i; q)$. We define the strategic advantage $\Delta(E)$ of the environment as the difference in prediction risk achieved by voluntary disclosure relative to mandatory disclosure: $\Delta(E) = R_V^*(q) - R_S^*(q)$, where $R_S^*(q)$ is the optimal strategic risk. A positive strategic advantage ($\Delta(E) > 0$) implies that voluntary disclosure is strictly beneficial to the receiver.

Under learning settings where the joint distribution $\mathcal{P}$ is unknown, the receiver must estimate the optimal default class and classifier from data. In the offline batch learning setup, the receiver has access to a training set of n i.i.d. samples and seeks to find an empirical default-classifier pair that minimizes the true strategic risk, bounding the generalization gap $R_S(\hat{i}, \hat{h}, q) - R_S(i^*, h^*, q)$ as a function of the sample size n. In the online sequential learning setup, the receiver updates the default class i_t and the classifier weights Θ_t over rounds $t = 1, \dots, T$ based only on the observed message $M_t \in \{x_t, \emptyset\}$ and the true label y_t (without observing the sender's private preference u_t), aiming to minimize the expected cumulative regret $\mathrm{Regret}(T) = \mathbb{E}[\sum_{t=1}^{T} R_S(i_t, \Theta_t; q)] - \min_{i,\Theta} \sum_{t=1}^{T} R_S(i, \Theta; q)$.

4.1 The Classification Decision Problem

Let $(U, X, Y) \sim \mathcal{P}$ be the joint random variable representing the sender's preference, features, and true labels, respectively. Senders and the receiver evaluate predictions using the 0-1 loss:

$$\ell(y, \hat{y}) = \mathbf{1}\{y \neq \hat{y}\}. \tag{26}$$

The receiver commits to a default prediction class $i \in \{1, \dots, K\}$ for empty messages, i.e., $\hat{y}(\emptyset) = i$. If features are revealed, the receiver deploys the Bayes classifier conditioned on the revealed subpopulation:

$$h_i(X) \in \operatorname*{arg\,max}_{c \in \{1,\dots,K\}} \Pr(Y = c \mid X,\, U \neq i). \tag{27}$$

Let $\pi_i = \Pr(U = i)$ denote the probability that the sender's preferred class is i. The Bayes risk on the subpopulation with preference $U \neq i$ is:

$$r_X^{(-i)} = \mathbb{E}\left[\mathbf{1}\{h_i(X) \neq Y\} \mid U \neq i\right]. \tag{28}$$

We first analyze the follower sender's best response to this policy.

Lemma 4.1 (Sender Best Response in Classification). *For a committed default class $i \in \{1, \dots, K\}$, the sender's utility-maximizing disclosure strategy is:*

$$D^*(i; U, X) = \mathbf{1}\{U \neq i\}. \tag{29}$$

That is, senders with preferred class i withhold features ($D = 0$), while senders with preferred class $U \neq i$ reveal features ($D = 1$).

The proof is postponed to Section 7.2.

Using the best-response policy, we derive the strategic risk for each committed default class i.

Lemma 4.2 (Strategic Risk for Default class i). *For a committed default class i, the receiver's strategic prediction risk under channel noise q is:*

$$R_S^i(q) = \Pr(Y \neq i) + (1 - \pi_i)(1 - q)\left(r_X^{(-i)} - \Pr(Y \neq i \mid U \neq i)\right). \tag{30}$$

The proof is postponed to Section 7.2.

Equation (30) shows that for any fixed default class i, the strategic risk is an affine function of the noise parameter q. This affine structure yields a strong geometric property at the endpoints of the noise interval.

Theorem 4.3 (Dominance of the Vanilla Best Constant). *Let $c^* \in \arg\min_i \Pr(Y \neq i)$ be the optimal default class under mandatory disclosure (the vanilla best constant). If c^* is the optimal default class at both $q = 1$ and $q = 0$, then it remains optimal for all $q \in [0, 1]$.*

The proof is postponed to Section 7.2.

By taking the lower envelope of the K affine strategic risk functions, we characterize the optimal strategic risk $R_S^*(q) = \min_i R_S^i(q)$.

Theorem 4.4 (Piecewise-Linear Partition of the Noise Interval). *The optimal strategic risk $R_S^*(q)$ is a concave, piecewise-linear function of q. The channel noise interval $[0, 1]$ is partitioned into at most K subintervals, on each of which a single default class is optimal.*

The proof is postponed to Section 7.2.

Remark 3. *This geometric structure allows the receiver to compute the optimal default regions efficiently. For each of the* $\binom{K}{2}$ *pairs of defaults, the receiver solves the linear equation* $R_S^i(q) = R_S^j(q)$*. The intersection points partition* $[0,1]$ *into a collection of active default regions.*

4.2 Offline Learning (Sample Complexity)

When the joint distribution $\mathcal{P}$ is unknown, the receiver must learn the optimal default class and classifier weights from n i.i.d. training samples $S = \{(u_j, x_j, y_j)\}_{j=1}^n$.

For each default class i, the receiver's model is parameterized by a weight matrix $\Theta_i \in \mathbb{R}^{K\times d}$ with $\|\Theta_i\|_\infty \leq C$. The score vector is $s = \Theta_i x$, and the classifier is:

$$h_{\Theta_i}(x) \in \underset{c\in\{1,\dots,K\}}{\arg\max}\,\langle \theta_{i,c}, x\rangle. \tag{31}$$

The hypothesis class is $\mathcal{H}_i = \{h_\Theta : \Theta \in \mathbb{R}^{K\times d},\ \|\Theta\|_\infty \leq C\}$.

The empirical strategic risk under default i and classifier h is:

$$\hat{R}_S^i(h,q) = \frac{1}{n}\sum_{j=1}^n \left[\mathbf{1}\{u_j = i\}\mathbf{1}\{i \neq y_j\} + \mathbf{1}\{u_j \neq i\}\left(q\mathbf{1}\{i \neq y_j\} + (1-q)\mathbf{1}\{h(x_j) \neq y_j\}\right)\right]. \tag{32}$$

The receiver deploys the joint selection empirical risk minimization algorithm detailed in Algorithm 3.

Algorithm 3 Strategic Classification Empirical Risk Minimization (SC-ERM)

1: **Input:** Training set $S = \{(u_j, x_j, y_j)\}_{j=1}^n$, class size K, channel erasure probability q, bounded domain parameter C.
2: **Define:** Hypothesis classes $\mathcal{H}_i = \{h_\Theta : \Theta \in \mathbb{R}^{K\times d},\ \|\Theta\|_\infty \leq C\}$ for all $i \in \{1,\dots,K\}$, where $h_\Theta(x) \in \arg\max_c \langle \theta_c, x\rangle$.
3: **for** each default class $i \in \{1,\dots,K\}$ **do**
4: Formulate the empirical strategic risk under default i:

$$\hat{R}_S^i(h,q) = \frac{1}{n}\sum_{j=1}^n \left[\mathbf{1}\{u_j = i\}\mathbf{1}\{i \neq y_j\} + \mathbf{1}\{u_j \neq i\}\left(q\mathbf{1}\{i \neq y_j\} + (1-q)\mathbf{1}\{h(x_j) \neq y_j\}\right)\right].$$

5: Compute the empirical strategic risk minimizer:

$$\hat{h}_i \in \underset{h\in\mathcal{H}_i}{\arg\min}\,\hat{R}_S^i(h,q).$$

6: **end for**
7: **Joint Selection:** Select the optimal default-predictor pair:

$$(\hat{i},\hat{h}) \in \underset{i\in\{1,\dots,K\},\, h=\hat{h}_i}{\arg\min}\ \hat{R}_S^i(h,q).$$

8: **Output:** Optimal default class $\hat{i}$ and strategic classifier $\hat{h}$.

We make the following assumption regarding the availability of data:

Assumption 1 (Nondegenerate Revealed Mass). *There exists a constant* $\gamma > 0$ *such that the probability of features being revealed is lower-bounded by* γ*, i.e.,* $\min_i \Pr(U \neq i) \geq \gamma$.

Under this assumption, we bound the sample size of the revealed subpopulation.

Lemma 4.5 (Effective Sample Size Concentration). *Let $\delta \in (0,1)$. With probability at least $1-\delta/2$ over the draw of S, the number of revealed samples $n_i = \sum_{j=1}^{n} \mathbf{1}\{u_j \neq i\}$ satisfies:*

$$n_i \geq \frac{\gamma n}{2} \quad \forall i \in \{1,\ldots,K\} \tag{33}$$

simultaneously, provided the training sample size n is sufficiently large.

The proof is postponed to Section 7.2.

To establish uniform generalization, we leverage the Natarajan dimension.

Lemma 4.6 (Natarajan's Sauer-Shelah Lemma). *Let $\mathcal{H}$ be a hypothesis class of functions from $\mathcal{X}$ to $\{1,\ldots,K\}$ with Natarajan dimension d_N. For any sample of size n, the number of distinct labelings $|\mathcal{H}|_S|$ is bounded by:*

$$|\mathcal{H}|_S| \leq \sum_{j=0}^{d_N} \binom{n}{j} \binom{K}{2}^j. \tag{34}$$

The proof is postponed to Section 7.2.

Using Natarajan's lemma and Rademacher complexity over discrete 0-1 losses, we establish fixed-default convergence.

Lemma 4.7 (Fixed-Default Uniform Convergence). *For any fixed default class i, with probability at least $1-\delta/K$:*

$$\sup_{h\in\mathcal{H}_i} |R_S^i(h,q) - \hat{R}_S^i(h,q)| \leq O\left(\sqrt{\frac{Kd\log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right). \tag{35}$$

The proof is postponed to Section 7.2.

By taking a union bound over the K default classes, we obtain simultaneous convergence.

Lemma 4.8 (Uniform Convergence Over All Defaults). *With probability at least $1-\delta$:*

$$\sup_{i\in\{1,\ldots,K\}} \sup_{h\in\mathcal{H}_i} |R_S^i(h,q) - \hat{R}_S^i(h,q)| \leq O\left(\sqrt{\frac{Kd\log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right). \tag{36}$$

The proof is postponed to Section 7.2.

This simultaneous generalization guarantees that the empirical optimal strategy achieves near-oracle excess risk.

Theorem 4.9 (Excess Risk of the Learned Strategy). *Let $(\hat{i}^*, \hat{h})$ be the strategy chosen by the joint selection learner. With probability at least $1-\delta$:*

$$R_{\hat{i}^*}^S(\hat{h},q) - \min_{i,h\in\mathcal{H}_i} R_S^i(h,q) \leq O\left(\sqrt{\frac{Kd\log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right). \tag{37}$$

The proof is postponed to Section 7.2.

4.3 Online Learning (Sequential Strategic Classification)

We now study online multiclass classification over rounds $t = 1,\ldots,T$. At each round t, the receiver commits to a default class $i_t \in \{1,\ldots,K\}$ and a predictor parameter matrix $\Theta_t \in \mathbb{R}^{K\times d}$. Nature then draws the private sender characteristics $(u_t, x_t, y_t) \sim \mathcal{P}$, and the sender discloses features if their preference u_t does not match the receiver's default i_t. Finally, the receiver observes the message $M_t = x_t$ if successfully disclosed (and not erased), and $M_t = \emptyset$ if withheld. The receiver predicts the class label using the default i_t on empty messages and the linear scoring model otherwise, suffering a 0-1 prediction loss.

The discontinuous 0-1 loss is non-convex. Furthermore, the receiver only observes features and label feedback for the committed default i_t (bandit feedback). We replace the 0-1 loss with the multiclass cross-entropy surrogate:

$$\phi(\Theta; x, y) = -\log\left(\frac{\exp(\langle\theta_y, x\rangle)}{\sum_{c=1}^{K}\exp(\langle\theta_c, x\rangle)}\right). \tag{38}$$

The surrogate strategic loss at round t is:

$$\tilde{\ell}_t(i, \Theta) = \mathbf{1}\{u_t = i\}\mathbf{1}\{i \neq y_t\} + \mathbf{1}\{u_t \neq i\}\phi(\Theta; x_t, y_t). \tag{39}$$

To learn both the discrete default classes and the continuous predictors, we propose the decoupled **EXP3 + OGD** algorithm detailed in Algorithm 4.

Algorithm 4 Online Multiclass Classification with Strategic Withholding

Initialize: For each $i \in \{1, \ldots, K\}$, initialize $\Theta_{1,i} \in \mathcal{W}$ and weight $w_{1,i} = 1$. Let $\eta > 0$ be the learning rate, and $\gamma \in (0, 1]$ be the exploration parameter.
for $t = 1, \ldots, T$ **do**
  Compute selection probabilities:
$$p_{t,i} = (1-\gamma)\frac{w_{t,i}}{\sum_{j=1}^{K} w_{t,j}} + \frac{\gamma}{K}.$$
  Sample default class $i_t \sim p_t$.
  Commit to default class i_t and predictor Θ_{t,i_t}.
  Observe message $M_t \in \{x_t, \emptyset\}$.
  if $M_t = \emptyset$ **then**
    Suffer loss $L_t = \mathbf{1}\{y_t \neq i_t\}$ (or use an unbiased proxy) and set $\Theta_{t+1,i_t} = \Theta_{t,i_t}$.
  else
    Predict $\hat{y}_t = \arg\max_c \langle\theta_{t,i_t,c}, x_t\rangle$ and observe true label y_t.
    Suffer surrogate loss $L_t = \phi(\Theta_{t,i_t}; x_t, y_t)$.
    Compute importance-weighted gradient:
$$\hat{g}_t = \frac{1}{p_{t,i_t}}\nabla\phi(\Theta_{t,i_t}; x_t, y_t).$$
    Update predictor weights: $\Theta_{t+1,i_t} = \Pi_{\mathcal{W}}\left(\Theta_{t,i_t} - \eta\hat{g}_t\right)$.
  end if
  For all unselected defaults $j \neq i_t$, keep weights unchanged: $\Theta_{t+1,j} = \Theta_{t,j}$.
  Construct unbiased loss estimate: $\hat{L}_{t,i} = \frac{L_t}{p_{t,i_t}}\mathbf{1}\{i = i_t\}$.
  Update EXP3 weights: $w_{t+1,i} = w_{t,i}\exp\left(-\frac{\gamma}{K}\hat{L}_{t,i}\right)$ for all $i \in \{1, \ldots, K\}$.
end for

We decompose the surrogate regret into the selection regret of EXP3 and the optimization regret of OGD.

Lemma 4.10 (EXP3 Regret Bound). *Let $L_{t,i} = \tilde{\ell}_t(i, \Theta_{t,i}) \in [0, B]$ be the strategic surrogate loss. Under Algorithm 4, the expected regret of the default selection against any fixed i^* is:*

$$\mathbb{E}\left[\sum_{t=1}^{T}\tilde{\ell}_t(i_t, \Theta_{t,i_t})\right] - \mathbb{E}\left[\sum_{t=1}^{T}\tilde{\ell}_t(i^*, \Theta_{t,i^*})\right] \leq B\gamma T + \frac{BK\ln K}{\gamma}. \tag{40}$$

The proof is postponed to Section 7.2.

Lemma 4.11 (Importance-Weighted OGD Regret). *Suppose the surrogate loss ϕ is G-Lipschitz in Θ, and $\mathcal{W}$ has Euclidean diameter D. For any fixed default i^* and predictor Θ^*, the OGD update satisfies:*

$$\mathbb{E}\left[\sum_{t=1}^{T}\tilde{\ell}_t(i^*, \Theta_{t,i^*})\right] - \sum_{t=1}^{T}\tilde{\ell}_t(i^*, \Theta^*) \leq \frac{D^2}{2\eta} + \frac{\eta G^2 KT}{2\gamma}. \tag{41}$$

The proof is postponed to Section 7.2.

By combining the selection and weights regret, we derive the overall expected strategic regret.

Theorem 4.12 (Classification Strategic Regret Guarantee). *By choosing the learning rate $\eta = \frac{D}{G}\sqrt{\frac{\gamma}{KT}}$ and the exploration rate $\gamma = \min\left\{1, \left(\frac{K\ln K}{T}\right)^{1/3}\right\}$, the expected strategic surrogate regret of Algorithm 4 satisfies:*

$$\mathbb{E}\left[\sum_{t=1}^{T}\tilde{\ell}_t(i_t, \Theta_{t,i_t})\right] - \min_{i,\Theta}\sum_{t=1}^{T}\tilde{\ell}_t(i, \Theta) \leq O\left(BT^{2/3}(K\ln K)^{1/3} + DGT^{2/3}K^{1/3}\right) = \tilde{O}(T^{2/3}). \tag{42}$$

The proof is postponed to Section 7.2.

Remark 4. *This result shows that online multiclass classification is learnable under strategic disclosure at a sublinear rate of $\tilde{O}(T^{2/3})$. Unlike the regression setting where the one-point bandit feedback costs a factor of d (dimension), the classification bandit setting restricts exploration to the K discrete default options, yielding a dimension-independent regret rate.*

5 Simulation Results and Empirical Validation

In this section, we present the empirical results of our simulations to validate the theoretical findings established in Sections 3 and 4. We begin by outlining the experimental setup, detailing the generative models for continuous and discrete prediction spaces, and then present the outcome of our parameter sweeps, offline learning rates, and online regret convergence.

5.1 Experimental Setup

To simulate realistic settings with asymmetric information and non-linear dependencies, we implement two distinct generative models:

5.1.1 Continuous Space (Regression Model). We generate the continuous variables $(X, Y, U) \in \mathbb{R}^3$ using a two-component Gaussian Mixture Model (GMM). This ensures that the data distribution is non-Gaussian, highlighting the generality of our framework:

(1) A latent component variable $z \in \{1, 2\}$ is drawn with equal probability.
(2) Conditioned on z, the variables are drawn from a multivariate Gaussian with mean vector μ_z and covariance matrix Σ_z. The means are shifted to create asymmetric regions: $\mu_1 = [1, 1, 1]^\top$ and $\mu_2 = [-1, -1, -1]^\top$.
(3) The covariance matrix is parameterized by the correlation coefficients ρ_{XY} (correlation between features and labels) and ρ_{XU} (correlation between features and sender preferences). The alignment between Y and U (beyond the features X) is controlled by a parameter $\gamma \in [0, 1]$,

set to $\gamma = 0.8$ throughout. The standard Gaussians are combined as:

$$X = \mu_{z,X} + Z_X, \tag{43}$$

$$Y = \mu_{z,Y} + \rho_{XY} Z_X + \sqrt{1-\rho_{XY}^2} Z_Y, \tag{44}$$

$$U = \mu_{z,U} + \rho_{XU} Z_X + \gamma\sqrt{1-\rho_{XU}^2} Z_Y + \sqrt{1-\gamma^2}\sqrt{1-\rho_{XU}^2} Z_U, \tag{45}$$

where $Z_X, Z_Y, Z_U \sim \mathcal{N}(0,1)$ are independent standard normal variables.

5.1.2 Discrete Space (Classification Model). For multiclass classification with K classes, the preference $U \in \{1,\ldots,K\}$ is drawn from a prior categorical distribution $\pi = [0.35, 0.4, 0.25]$ (for $K = 3$). Conditioned on the preference $U = i$:

(1) The features $X \in \mathbb{R}^2$ are drawn from a class-conditional Gaussian:

$$X \mid U = i \sim \mathcal{N}(\mu_i, \sigma^2 I), \tag{46}$$

where $\mu_1 = [-1,-1]^\top$, $\mu_2 = [0, 1.5]^\top$, and $\mu_3 = [1,-1]^\top$ represent centroids distributed on a circle, and the class variance is $\sigma^2 = 0.5$.

(2) The true label $Y \in \{1,\ldots,K\}$ is drawn such that the sender's preference is aligned with Y with probability $1-\alpha$, and is uniformly distributed over the remaining $K-1$ classes with probability α (alignment noise):

$$\Pr(Y = j \mid U = i) = (1-\alpha)\mathbf{1}\{j = i\} + \frac{\alpha}{K-1}\mathbf{1}\{j \neq i\}. \tag{47}$$

—

5.2 Decision Sweeps

We first validate our game-theoretic foundations by sweeping the environment parameters and plotting the resulting strategic advantage $\Delta^* = R_V^* - R_S^*$ achieved by the receiver under voluntary disclosure.

5.2.1 Regression Decision Sweep. Figure 1 displays the strategic advantage heatmap for the regression game under a fixed channel noise $q = 0.3$. We sweep ρ_{XY} and ρ_{XU} on a 25×25 grid. * **Observations:** A positive strategic advantage (indicated by red regions) is attained when the feature-label correlation ρ_{XY} and the feature-preference correlation ρ_{XU} have the same sign. In this regime, the sender's disclosure incentive aligns with the receiver's predictive benefit: the sender voluntarily reveals features when they are large in the direction of the label, which coincides with the instances where features are highly predictive. Conversely, when the correlations have opposite signs, voluntary disclosure degrades accuracy compared to OLS Omit baselines (blue regions). * **Sufficient Condition Validation:** The dashed black contour shows the boundary where the sufficient condition (Theorem 3.4) holds. As predicted, this region is a subset of the true positive strategic advantage region, confirming that the condition is sufficient but conservative due to the Cauchy-Schwarz approximation of the preference bias.

5.2.2 Classification Decision Sweep. Figure 2 (left) plots the strategic risks $R_S^i(q)$ for each of the $K = 3$ default classes alongside the optimal strategic risk envelope $R_S^*(q) = \min_i R_S^i(q)$, for a fixed alignment noise $\alpha = 0.15$. * **Piecewise-Linear Structure:** In accordance with Lemma 4.2, each individual risk line is affine in q. The lower envelope $R_S^*(q)$ is concave and piecewise-linear, partitioned into three linear segments separated by transition points near $q \approx 0.35$ and $q \approx 0.70$. This confirms Theorem 4.4: the optimal default class changes from class 2 to class 1, and finally to class 3 as the channel noise increases, partitioning the noise interval. * **Advantage Heatmap:** Figure 2 (right) sweeps the channel noise q and alignment noise α. The strategic advantage is

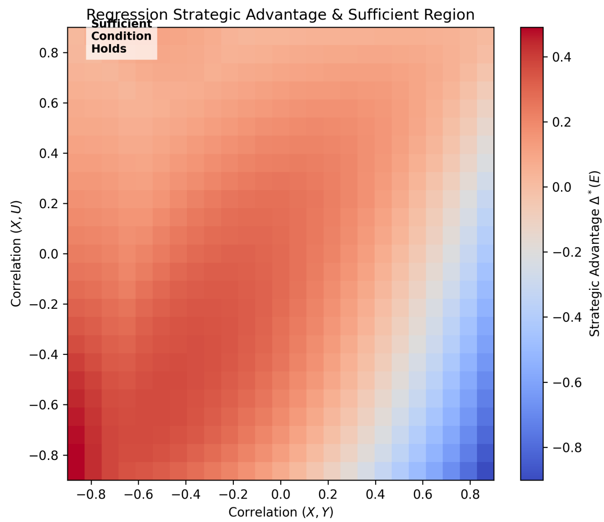

Fig. 1. Regression Strategic Advantage $\Delta^*(E)$ as a function of feature-label correlation ρ_{XY} and feature-preference correlation ρ_{XU} (under channel noise $q = 0.3$, employing a coolwarm colormap). The dashed black contour delineates the boundary of the sufficient condition (Theorem 3.4).

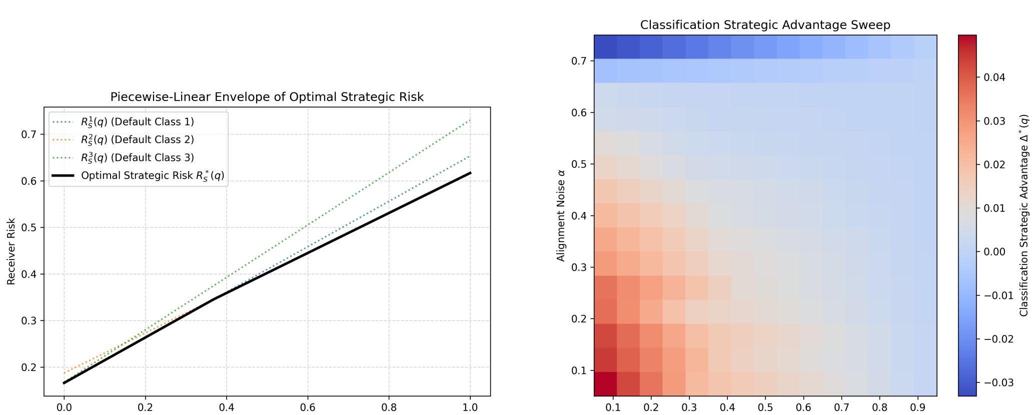

Fig. 2. Classification decision results. Left: Piecewise-linear envelope of strategic risks for different defaults i, validating Theorem 4.4; Right: Heatmap of strategic advantage $\Delta^*(q)$ over noise q and alignment noise α (consistent coolwarm colormap).

largest when the alignment noise α is low (indicating high alignment between preference U and label Y) and the channel noise q is small (ensuring features are successfully transmitted). As the alignment noise α increases beyond 0.5, the preference becomes decoupled from the target class, and the strategic advantage vanishes.

—

5.3 Offline Generalization Gaps

Next, we evaluate the offline learning curves by performing multiple experiments with different training sample sizes $n \in \{50, 100, 200, 500, 1000, 2000, 4000\}$. We explicitly track the excess strategic risk $R_S(\hat{b}_n) - R_S(b^*)$ (for regression) and $R_S(\hat{\Theta}_n) - R_S(\Theta^*)$ (for multiclass classification) for each distinct training size n, represented as discrete points in our scatter plots, and averaged over 15 independent runs.

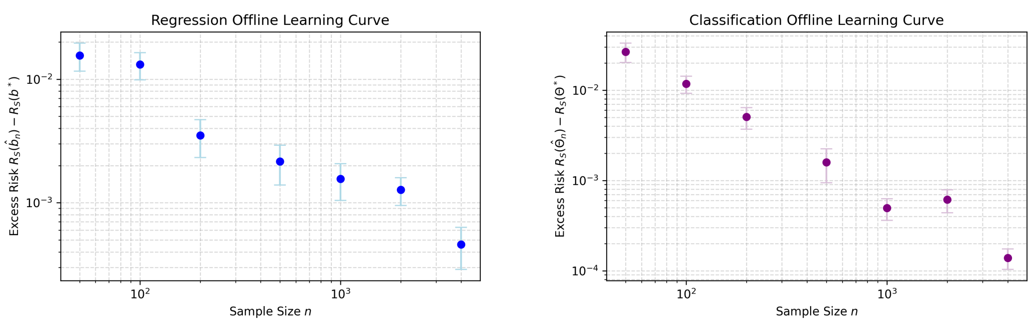

Fig. 3. Offline learning scatter plots showing the decay of the excess strategic risk on a log-log scale for varying training sizes n. Left: Regression excess risk converging at a rate matching the $O(d^3/\sqrt{n})$ bound; Right: Multiclass classification excess risk converging at a rate matching the $O(\sqrt{Kd/n})$ Natarajan dimension bound.

* **Regression Convergence:** Figure 3 (left) shows the convergence rate of the strategic empirical risk minimizer (ERM) on a log-log scale. The excess risk decays rapidly, demonstrating that a sample size of $n \geq 1000$ is sufficient to achieve an excess strategic prediction risk of less than 0.05. The decay matches the theoretical $O(n^{-0.5})$ generalization rate proved in Theorem 3.9. * **Classification Convergence:** Figure 3 (right) validates the sample complexity of the joint selection algorithm in classification. The excess risk converges to zero as n approaches 4000, validating Lemma 4.8 and Theorem 4.9. The convergence rate matches the sublinear rate $O(n^{-0.5})$ governed by the Natarajan dimension of multiclass linear boundaries.

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5.4 Online Regret Convergence

Finally, we analyze the online sequential learning setting under bandit feedback, running the algorithms over $T = 3000$ steps.

* **Regression Bandit OGD:** Figure 4 (left) displays a scatter plot of the cumulative regret of the one-point bandit gradient estimator. For both dimensions $d = 1$ and $d = 3$, the regret grows sublinearly, tightly bounded by their respective dimension-dependent $O(d\sqrt{t})$ theoretical benchmarks. Comparing the two, we clearly observe that the higher dimension $d = 3$ incurs proportionally larger cumulative regret. This is due to the d/δ_t scaling in the one-point spherical gradient estimate, matching Theorem 3.11. * **Classification EXP3 + OGD:** Figure 4 (right) presents a scatter plot of the expected cumulative regret of our decoupled EXP3+OGD algorithm (Algorithm 1) for $K = 2$ and $K = 3$ classes. The regret grows sublinearly, conforming to the distinct K-dependent $O((K \ln K)^{1/3} t^{2/3})$ reference curves proved in Theorem 4.12. The curve for $K = 3$ classes exhibits a slower convergence rate and higher regret than $K = 2$ classes, as the meta-algorithm must explore a larger discrete choice space (EXP3 arms) to identify the optimal default class.

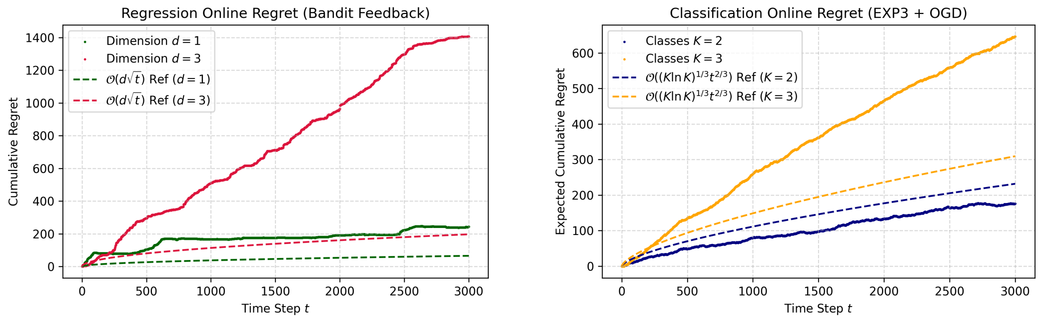

Fig. 4. Online expected cumulative regret scatter plots under bandit feedback. Left: Regression regret under bandit OGD using one-point gradient estimates for $d = 1$ and $d = 3$, showing $O(d\sqrt{t})$ sublinear growth with dimension-dependent reference benchmarks; Right: Classification regret under EXP3+OGD for $K = 2$ and $K = 3$, confirming the sublinear $O((K \ln K)^{1/3} t^{2/3})$ regret bound with class-dependent benchmarks.

6 Conclusion and Future Work

In this paper, we have presented a unified framework for machine learning under strategic information disclosure, specifically focusing on the problem of strategic feature withholding. By modeling the interaction as a Stackelberg game between a data provider (the sender) and a decision-maker (the receiver) connected through a noisy erasure channel, we analyzed the optimal strategies and learning dynamics across two fundamental prediction spaces—continuous labels (regression) and discrete labels (multiclass classification)—and three sequential regimes: the decision-theoretic game, offline batch learning, and online sequential learning.

For the regression setting, we established necessary and sufficient conditions for voluntary disclosure to benefit the receiver, derived uniform generalization bounds of $O(d^3/\sqrt{n})$ for the offline empirical risk minimizer, and proved sublinear online regret bounds of $O(d\sqrt{T})$ under bandit feedback via a spherical one-point gradient estimator. For the multiclass classification setting, we proved that the optimal strategic risk forms a concave piecewise-linear envelope over the noise space, derived excess risk bounds of $O(\sqrt{Kd \log n/n})$ via the Natarajan dimension, and designed a decoupled EXP3+OGD algorithm achieving $\tilde{O}(T^{2/3})$ expected regret under bandit feedback. All theoretical results were validated through comprehensive GMM-based simulations, confirming the strategic advantage boundaries, piecewise-linear risk structures, and convergence rates.

6.1 Limitations

While our framework yields a rich set of theoretical guarantees, several important limitations remain:

- **All-or-Nothing Disclosure.** Throughout this work, we assume senders can only make a binary choice: reveal the *entire* feature vector X or withhold it completely ($\emptyset$). This is a significant modeling restriction. In practice, agents may selectively disclose *subsets* of features, leading to a combinatorially larger action space that our current analysis does not handle.
- **Gap Between Necessary and Sufficient Conditions.** For regression, our sufficient condition (Theorem 3.4) and necessary condition (Theorem 3.5) do not meet: there exists a region of the parameter space where the true strategic advantage is positive but our sufficient condition does not confirm it. Tightening this gap remains an open problem.

- **Local Optimality in Online Regression.** Because the strategic risk is non-convex in the receiver's parameters b, our online gradient-based algorithms only converge to a *local* minimizer within a convex basin, not a global optimum. This local regret guarantee may be insufficient in environments with multiple local minima.
- **Known Noise Parameter q.** Our offline learning algorithms assume the channel erasure probability q is known to the receiver. In practice, q must be estimated from data, adding an additional layer of statistical complexity not captured by our current bounds.
- **Nondegeneracy Assumption.** For the classification offline bounds, we require Assumption 1: each default class receives a nontrivial fraction $\gamma > 0$ of revealed samples. When the prior is highly concentrated on a single class, this assumption fails and our analysis breaks down.
- **Slower Online Regret for Classification.** The $\tilde{O}(T^{2/3})$ regret for classification (compared to $O(\sqrt{T})$ for regression) reflects the inherent cost of the discrete explore-exploit trade-off in EXP3. Bridging this gap using improved bandit algorithms is an open direction.

6.2 Future Directions

There are several promising avenues for extending this work:

- **Selective Feature Withholding.** Relaxing the all-or-nothing model to allow agents to reveal or withhold arbitrary subsets of features. This would introduce combinatorial challenges to the receiver's model selection and requires new tools from set-function optimization.
- **Non-linear and Neural Representations.** Generalizing our generalization bounds and online regret results to non-linear hypothesis classes (e.g., kernel methods, neural networks) using neural tangent kernels or Rademacher complexities of deep models.
- **Multi-Sender Environments.** Extending the framework to settings with multiple heterogeneous senders with potentially competing or cooperative disclosure strategies. This connects to mechanism design and information aggregation theory.
- **Non-Stationary and Adversarial Environments.** Analyzing online learning under non-stationary distributions or adaptive adversaries, where the joint distribution $\mathcal{P}$ or the channel noise q drifts over time, requiring tracking regret bounds.
- **Partial Feature Observability.** Studying settings where the receiver can *query* a subset of features at a cost, creating an active learning variant of the strategic disclosure game.

7 Postponed Proofs

This section collects the complete formal proofs for all theorems and lemmas referenced in the main text, organized by result domain (regression and classification). Each proof is presented with a clearly labeled environment indicating the specific result being proven, allowing readers to easily locate and reference individual proofs.

7.1 Proofs for Regression (Section 3)

The following proofs establish the game-theoretic foundations, offline generalization bounds, and online regret guarantees for the continuous regression setting.

Proof of Lemma 3.1: Vanilla Risk Decomposition.

Vanilla Risk Decomposition. Under mandatory disclosure, the sender's action is fixed to always reveal ($D \equiv 1$). The feature vector X is successfully transmitted with probability $1 - q$ and is erased by the channel with probability q. If erasure occurs, the receiver predicts the default value 0 and suffers quadratic loss Y^2. If the transmission succeeds, the receiver predicts $b^\top X$ and suffers

$(Y - b^\top X)^2$. By definition, the vanilla risk is:

$$R_{\mathrm{V}}(b, q) = q\mathbb{E}[Y^2] + (1-q)\mathbb{E}[(Y - b^\top X)^2]. \tag{48}$$

Since $\mathbb{E}[Y] = 0$, we have $\mathbb{E}[Y^2] = \mathrm{Var}(Y)$.

To decompose the second term, we add and subtract the OLS predictor $b_{\mathrm{V}}^\top X$:

$$\mathbb{E}[(Y - b^\top X)^2] = \mathbb{E}\left[\left((Y - b_{\mathrm{V}}^\top X) + (b_{\mathrm{V}} - b)^\top X\right)^2\right]. \tag{49}$$

Expanding the square yields:

$$\mathbb{E}[(Y - b^\top X)^2] = \mathbb{E}[(Y - b_{\mathrm{V}}^\top X)^2] + \mathbb{E}[((b_{\mathrm{V}} - b)^\top X)^2] + 2\mathbb{E}[(Y - b_{\mathrm{V}}^\top X)(b_{\mathrm{V}} - b)^\top X]. \tag{50}$$

By the orthogonality principle of linear least squares, the error $Y - b_{\mathrm{V}}^\top X$ is orthogonal to the feature space $\mathcal{X}$, meaning $\mathbb{E}[(Y - b_{\mathrm{V}}^\top X)X] = 0$. Consequently, the cross-product term is zero:

$$2(b_{\mathrm{V}} - b)^\top \mathbb{E}[(Y - b_{\mathrm{V}}^\top X)X] = 0. \tag{51}$$

Since $\mathbb{E}[Y] = 0$ and $\mathbb{E}[b_{\mathrm{V}}^\top X] = b_{\mathrm{V}}^\top \mathbb{E}[X] = 0$, we have:

$$\mathbb{E}[(Y - b_{\mathrm{V}}^\top X)^2] = \mathbb{E}[Y^2] - \mathbb{E}[(b_{\mathrm{V}}^\top X)^2] = \mathrm{Var}(Y) - \mathrm{Var}(b_{\mathrm{V}}^\top X), \tag{52}$$

and

$$\mathbb{E}[((b_{\mathrm{V}} - b)^\top X)^2] = \mathrm{Var}((b_{\mathrm{V}} - b)^\top X) = \mathrm{Var}((b - b_{\mathrm{V}})^\top X). \tag{53}$$

Substituting these terms back into the risk formula gives:

$$\begin{aligned} R_{\mathrm{V}}(b, q) &= q\mathrm{Var}(Y) + (1-q)\left(\mathrm{Var}(Y) - \mathrm{Var}(b_{\mathrm{V}}^\top X) + \mathrm{Var}((b - b_{\mathrm{V}})^\top X)\right) \\ &= \mathrm{Var}(Y) - (1-q)\mathrm{Var}(b_{\mathrm{V}}^\top X) + (1-q)\mathrm{Var}((b - b_{\mathrm{V}})^\top X), \end{aligned} \tag{54}$$

which matches (7).

The vanilla risk is minimized when the non-negative term $\mathrm{Var}((b - b_{\mathrm{V}})^\top X)$ is zero. Assuming $\mathrm{Var}(X) \succ 0$, this occurs uniquely at $b = b_{\mathrm{V}}$, yielding the optimal vanilla risk:

$$R_{\mathrm{V}}^*(q) = \mathrm{Var}(Y) - (1-q)\mathrm{Var}(b_{\mathrm{V}}^\top X). \tag{55}$$

□

Proof of Lemma 3.2: Sender Best Response.

Sender Best Response. Consider a sender with preference realization u and features x under a committed receiver predictor parameter b. If the sender chooses to withhold ($D = 0$), the receiver receives $M = \emptyset$, predicts 0, and the sender suffers loss:

$$\ell_S(0, u) = u^2. \tag{56}$$

If the sender chooses to disclose ($D = 1$), the feature is successfully transmitted with probability $1 - q$ (resulting in prediction $b^\top x$ and loss $(b^\top x - u)^2$) and is erased with probability q (resulting in prediction 0 and loss u^2). The sender's expected loss from disclosing is:

$$\mathbb{E}_\epsilon[\ell_S(\hat{Y}, u)] = (1-q)(b^\top x - u)^2 + qu^2. \tag{57}$$

The sender prefers disclosure if and only if:

$$u^2 > (1-q)(b^\top x - u)^2 + qu^2. \tag{58}$$

Since $q \in (0, 1)$, we divide by the positive constant $1 - q > 0$ to get:

$$u^2 > (u - b^\top x)^2. \tag{59}$$

Expanding the right-hand side gives:

$$u^2 > u^2 - 2ub^\top x + (b^\top x)^2 \iff (2u - b^\top x)b^\top x > 0. \tag{60}$$

Recalling the definition of $F_b(U, X)$ in (5), this simplifies to $F_b(u, x) > 0$. We resolve ties by withholding, which completes the proof. □

Proof of Lemma 3.3: Strategic Risk Formula.

Strategic Risk Formula. Under voluntary disclosure, Lemma 3.2 dictates that the sender reveals features if and only if $F_b(U, X) > 0$. Let $I_t = \mathbf{1}\{F_b(U, X) > 0\}$. If $I_t = 0$, the sender withholds features, and the receiver predicts 0, suffering loss Y^2. If $I_t = 1$, the sender attempts revelation. With probability q, the channel erases the features, leading to prediction 0 and loss Y^2. With probability $1 - q$, features are successfully received, resulting in prediction $b^\top X$ and loss $(Y - b^\top X)^2$.

The strategic risk is:

$$R_S(b, q) = \mathbb{E}\left[(1 - I_t)Y^2 + I_t\left(qY^2 + (1 - q)(Y - b^\top X)^2\right)\right]. \tag{61}$$

Grouping the terms containing Y^2 yields:

$$\begin{aligned} R_S(b, q) &= \mathbb{E}\left[Y^2 - (1 - q)I_tY^2 + (1 - q)I_t(Y - b^\top X)^2\right] \\ &= \mathbb{E}\left[Y^2\right] - (1 - q)\mathbb{E}\left[I_t\left(Y^2 - (Y - b^\top X)^2\right)\right]. \end{aligned} \tag{62}$$

Since $\mathbb{E}[Y] = 0$, we have $\mathbb{E}[Y^2] = \mathrm{Var}(Y)$. Expanding the term $Y^2 - (Y - b^\top X)^2$ gives:

$$Y^2 - (Y^2 - 2Yb^\top X + (b^\top X)^2) = (2Y - b^\top X)b^\top X = G_b(X, Y), \tag{63}$$

where $G_b(X, Y)$ is defined as in (6). Thus, the expression reduces to:

$$R_S(b, q) = \mathrm{Var}(Y) - (1 - q)\mathbb{E}\left[G_b(X, Y)\mathbf{1}\{F_b(U, X) > 0\}\right], \tag{64}$$

which is the stated equation (10). □

Proof of Theorem 3.4: Sufficient Condition for Strategic Advantage.

Sufficient Condition for Strategic Advantage. Let $\Delta(b, q) = R_V^*(q) - R_S(b, q)$. By definition, if there exists $b \in \mathbb{R}^d$ such that $\Delta(b, q) > 0$, then $\Delta(E) > 0$. Using the risk equations from Lemmas 3.1 and 3.3, the risk difference is:

$$R_S(b, q) - R_V^*(q) = (1 - q)\left[\mathrm{Var}(b_V^\top X) - \mathbb{E}\left[G_b(X, Y)\mathbf{1}\{F_b(U, X) > 0\}\right]\right]. \tag{65}$$

We rewrite the expectation as:

$$\mathbb{E}\left[G_b(X, Y)\mathbf{1}\{F_b(U, X) > 0\}\right] = \mathbb{E}[G_b(X, Y)] - \mathbb{E}\left[G_b(X, Y)\mathbf{1}\{F_b(U, X) \le 0\}\right]. \tag{66}$$

Note that $\mathbb{E}[G_b(X, Y)] = \mathbb{E}[(2Y - b^\top X)b^\top X] = 2b^\top\mathbb{E}[XY] - b^\top\mathbb{E}[XX^\top]b = 2b^\top\mathrm{Var}(X)b_V - \mathrm{Var}(b^\top X)$. We can rewrite this using the algebraic variance identity:

$$\mathrm{Var}((b - b_V)^\top X) = \mathrm{Var}(b^\top X) - 2b^\top\mathrm{Var}(X)b_V + \mathrm{Var}(b_V^\top X), \tag{67}$$

which implies:

$$\mathbb{E}[G_b(X, Y)] = \mathrm{Var}(b_V^\top X) - \mathrm{Var}((b - b_V)^\top X). \tag{68}$$

Thus, the risk difference is:

$$R_S(b, q) - R_V^*(q) = (1 - q)\left[\mathrm{Var}((b - b_V)^\top X) + \mathbb{E}\left[G_b(X, Y)\mathbf{1}\{F_b(U, X) \le 0\}\right]\right]. \tag{69}$$

For this difference to be strictly negative (signifying positive strategic advantage), we require the bracketed term to be negative.

We decompose the expectation term based on the sign of G_b:

$$\mathbb{E}\left[G_b\mathbf{1}\{F_b \le 0\}\right] = \mathbb{E}\left[G_b\mathbf{1}\{G_b \le 0\}\right] + \mathbb{E}\left[G_b\left(\mathbf{1}\{F_b \le 0\} - \mathbf{1}\{G_b \le 0\}\right)\right]. \tag{70}$$

For the first term, we write:

$$\mathbb{E}\left[G_b \mathbf{1}\{G_b \leq 0\}\right] = \frac{\mathbb{E}[G_b] - \mathbb{E}[|G_b|]}{2} = \frac{\mathrm{Var}(b_{\mathrm{V}}^{\top}X) - \mathrm{Var}((b - b_{\mathrm{V}})^{\top}X) - \mathbb{E}[|G_b|]}{2}. \tag{71}$$

For the second term, the integrand is non-zero only when F_b and G_b have opposite signs. On this mismatch event, we have $|G_b(X, Y)| \leq |G_b(X, Y) - F_b(U, X)|$. Using the definitions in (5) and (6), we have:

$$G_b(X, Y) - F_b(U, X) = 2(Y - U)b^{\top}X. \tag{72}$$

Therefore,

$$\mathbb{E}\left[G_b \left(\mathbf{1}\{F_b \leq 0\} - \mathbf{1}\{G_b \leq 0\}\right)\right] \leq \mathbb{E}\left[|G_b|\mathbf{1}\{\mathrm{sign}(F_b) \neq \mathrm{sign}(G_b)\}\right] \leq 2\mathbb{E}\left[|Y - U||b^{\top}X|\right]. \tag{73}$$

Applying the Cauchy-Schwarz inequality gives:

$$2\mathbb{E}\left[|Y - U||b^{\top}X|\right] \leq 2\|U - Y\|_2\sqrt{\mathrm{Var}(b^{\top}X)}. \tag{74}$$

Combining these components, we get:

$$\mathbb{E}\left[G_b \mathbf{1}\{F_b \leq 0\}\right] \leq \frac{\mathrm{Var}(b_{\mathrm{V}}^{\top}X) - \mathrm{Var}((b - b_{\mathrm{V}})^{\top}X) - \mathbb{E}[|G_b|]}{2} + 2\|U - Y\|_2\sqrt{\mathrm{Var}(b^{\top}X)}. \tag{75}$$

Substitute this inequality back into the risk difference:

$$R_{\mathrm{S}}(b, q) - R_{\mathrm{V}}^{*}(q) \leq (1 - q)\left[\frac{\mathrm{Var}(b_{\mathrm{V}}^{\top}X) + \mathrm{Var}((b - b_{\mathrm{V}})^{\top}X) - \mathbb{E}[|G_b|]}{2} + 2\|U - Y\|_2\sqrt{\mathrm{Var}(b^{\top}X)}\right]. \tag{76}$$

To guarantee that $R_{\mathrm{S}}(b, q) - R_{\mathrm{V}}^{*}(q) < 0$, it is sufficient for the term in the brackets to be strictly negative:

$$2\|U - Y\|_2\sqrt{\mathrm{Var}(b^{\top}X)} < \frac{\mathbb{E}[|G_b|] - \mathrm{Var}(b_{\mathrm{V}}^{\top}X) - \mathrm{Var}((b - b_{\mathrm{V}})^{\top}X)}{2}. \tag{77}$$

Rearranging this inequality yields (12). □

Proof of Theorem 3.5: Necessary Condition for Strategic Advantage.

Necessary Condition for Strategic Advantage. We analyze the risk difference under voluntary disclosure:

$$R_{\mathrm{S}}(b, q) - R_{\mathrm{V}}^{*}(q) = (1 - q)\left[\mathrm{Var}((b - b_{\mathrm{V}})^{\top}X) + \mathbb{E}\left[G_b(X, Y)\mathbf{1}\{F_b(U, X) \leq 0\}\right]\right]. \tag{78}$$

Pointwise, for any realizations, we have:

$$G_b \mathbf{1}\{F_b \leq 0\} \geq G_b \mathbf{1}\{G_b \leq 0\}. \tag{79}$$

This is established by cases:

- If $G_b \leq 0$, then the right-hand side is G_b. Since $G_b \mathbf{1}\{F_b \leq 0\}$ is either G_b (if $F_b \leq 0$) or 0 (if $F_b > 0$), and $0 \geq G_b$, the inequality holds.
- If $G_b > 0$, the right-hand side is 0. The left-hand side is either $G_b > 0$ or 0, which is always non-negative, so the inequality holds.

Taking expectations on both sides yields:

$$\mathbb{E}\left[G_b \mathbf{1}\{F_b \leq 0\}\right] \geq \mathbb{E}\left[G_b \mathbf{1}\{G_b \leq 0\}\right] = \frac{\mathrm{Var}(b_{\mathrm{V}}^{\top}X) - \mathrm{Var}((b - b_{\mathrm{V}})^{\top}X) - \mathbb{E}[|G_b|]}{2}. \tag{80}$$

Substitute this lower bound into the risk difference:

$$R_{\mathrm{S}}(b,q) - R^*_{\mathrm{V}}(q) \geq (1-q)\left[\mathrm{Var}((b-b_{\mathrm{V}})^\top X) + \frac{\mathrm{Var}(b_{\mathrm{V}}^\top X) - \mathrm{Var}((b-b_{\mathrm{V}})^\top X) - \mathbb{E}[|G_b|]}{2}\right]$$
$$= \frac{1-q}{2}\left[\mathrm{Var}(b_{\mathrm{V}}^\top X) + \mathrm{Var}((b-b_{\mathrm{V}})^\top X) - \mathbb{E}[|G_b|]\right]. \tag{81}$$

Now, suppose that for all $b \in \mathbb{R}^d$, the necessary condition (13) fails, meaning:

$$\mathbb{E}[|G_b(X,Y)|] - \mathrm{Var}(b_{\mathrm{V}}^\top X) - \mathrm{Var}((b-b_{\mathrm{V}})^\top X) \leq 0 \quad \forall b \in \mathbb{R}^d. \tag{82}$$

This implies that the term inside the brackets is always non-negative:

$$\mathrm{Var}(b_{\mathrm{V}}^\top X) + \mathrm{Var}((b-b_{\mathrm{V}})^\top X) - \mathbb{E}[|G_b(X,Y)|] \geq 0 \quad \forall b \in \mathbb{R}^d. \tag{83}$$

Since $1-q > 0$, this guarantees $R_{\mathrm{S}}(b,q) - R^*_{\mathrm{V}}(q) \geq 0$ for all b, which means $\Delta(E) \leq 0$. By contrapositive, if $\Delta(E) > 0$, there must exist some $b \in \mathbb{R}^d$ that satisfies the necessary condition (13). □

Proof of Lemma 3.6: Uniform Vanilla Generalization.

Uniform Vanilla Generalization. Let $L(b) = \mathbb{E}[(Y - b^\top X)^2]$ and $\hat{L}(b) = \frac{1}{n}\sum_{i=1}^n (y_i - b^\top x_i)^2$. The difference between the true and empirical vanilla risks is:

$$R_{\mathrm{V}}(b,q) - \hat{R}_{\mathrm{V}}(b,q) = q\mathrm{Var}(Y) + (1-q)L(b) - \left(q\frac{1}{n}\sum_{i=1}^n y_i^2 + (1-q)\hat{L}(b)\right)$$
$$= q\left(\mathrm{Var}(Y) - \frac{1}{n}\sum_{i=1}^n y_i^2\right) + (1-q)(L(b) - \hat{L}(b)). \tag{84}$$

Using the triangle inequality, we bound this difference uniformly:

$$\sup_{b\in\mathcal{W}} |R_{\mathrm{V}}(b,q) - \hat{R}_{\mathrm{V}}(b,q)| \leq \left|\mathrm{Var}(Y) - \frac{1}{n}\sum_{i=1}^n y_i^2\right| + \sup_{b\in\mathcal{W}} |L(b) - \hat{L}(b)|. \tag{85}$$

We allocate a failure probability of $\delta/2$ to each of these two terms using a union bound.

For the first term, since $Y \in [-C, C]$, the variable Y^2 lies in the interval $[0, C^2]$. Applying Hoeffding's concentration inequality yields:

$$\Pr\left(\left|\mathrm{Var}(Y) - \frac{1}{n}\sum_{i=1}^n y_i^2\right| > \epsilon_1\right) \leq 2\exp\left(-\frac{2n\epsilon_1^2}{C^4}\right). \tag{86}$$

Setting the probability bound equal to $\delta/2$ and solving for ϵ_1 gives:

$$\left|\mathrm{Var}(Y) - \frac{1}{n}\sum_{i=1}^n y_i^2\right| \leq C^2\sqrt{\frac{\log(4/\delta)}{2n}} = \mathcal{O}\left(C^2\sqrt{\frac{\log(1/\delta)}{n}}\right) \tag{87}$$

with probability at least $1-\delta/2$.

For the second term, we apply the Rademacher generalization bound. Let $\mathcal{F} = \{(x,y) \mapsto (y - b^\top x)^2 : b \in \mathcal{W}\}$ be the loss function class. Under the boundedness assumptions, $y_i \in [-C, C]$ and $b^\top x_i \in [-dC^2, dC^2]$ (since $\|b\|_\infty \leq C$ and $\|x_i\|_\infty \leq C$). The argument of the quadratic loss is bounded by $C + dC^2$. The derivative of the loss function $\ell(z) = (y-z)^2$ with respect to $z = b^\top x$ is bounded by $2(C + dC^2) = \mathcal{O}(dC^2)$. Thus, the loss function is $\mathcal{O}(dC^2)$-Lipschitz.

Let $\mathcal{H} = \{(x, y) \mapsto y - b^\top x : b \in \mathcal{W}\}$ be the linear function class. The Rademacher complexity of $\mathcal{H}$ satisfies $\hat{\mathcal{R}}(\mathcal{H}; S) = O\left(\frac{dC^2}{\sqrt{n}}\right)$. Using Talagrand's contraction lemma, the Rademacher complexity of the composed class $\mathcal{F} = \ell \circ \mathcal{H}$ is bounded by:

$$\hat{\mathcal{R}}(\mathcal{F}; S) \le O(dC^2)\hat{\mathcal{R}}(\mathcal{H}; S) = O\left(\frac{d^2C^4}{\sqrt{n}}\right). \tag{88}$$

Applying the Rademacher generalization theorem, with probability at least $1 - \delta/2$, we have:

$$\sup_{b \in \mathcal{W}} |L(b) - \hat{L}(b)| \le 2\hat{\mathcal{R}}(\mathcal{F}; S) + 4C^2\sqrt{\frac{\log(4/\delta)}{2n}} = O\left(\frac{d^2C^4}{\sqrt{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\right). \tag{89}$$

Summing these two uniform bounds completes the proof. □

Proof of Lemma 3.7: Uniform Strategic Generalization.

Uniform Strategic Generalization. Let $h_b(u, x, y) = G_b(x, y)\mathbf{1}\{F_b(u, x) > 0\}$. The strategic risk and empirical risk are:

$$R_S(b, q) = \mathrm{Var}(Y) - (1 - q)\mathbb{E}[h_b(U, X, Y)], \tag{90}$$

$$\hat{R}_S(b, q) = \frac{1}{n}\sum_{i=1}^{n} y_i^2 - (1 - q)\frac{1}{n}\sum_{i=1}^{n} h_b(u_i, x_i, y_i). \tag{91}$$

Applying the triangle inequality:

$$\sup_{b \in \mathcal{W}} |R_S(b, q) - \hat{R}_S(b, q)| \le \left|\mathrm{Var}(Y) - \frac{1}{n}\sum_{i=1}^{n} y_i^2\right| + \sup_{b \in \mathcal{W}} |\mathbb{E}[h_b] - \hat{\mathbb{E}}[h_b]|. \tag{92}$$

The variance term is identical to Lemma 3.6 and is bounded by $O\left(C^2\sqrt{\log(1/\delta)/n}\right)$ with probability at least $1 - \delta/2$.

For the second term, we analyze the complexity of the product class $\mathcal{F} \cdot \mathcal{G}$, where:

$$\mathcal{F} = \{(u, x) \mapsto \mathbf{1}\{F_b(u, x) > 0\} : b \in \mathcal{W}\}, \tag{93}$$

and

$$\mathcal{G} = \{(x, y) \mapsto G_b(x, y) : b \in \mathcal{W}\}. \tag{94}$$

For any product class, the empirical Rademacher complexity is bounded by:

$$\hat{\mathcal{R}}(\mathcal{F} \cdot \mathcal{G}; S) \le C_{\mathcal{G}}\hat{\mathcal{R}}(\mathcal{F}; S) + C_{\mathcal{F}}\hat{\mathcal{R}}(\mathcal{G}; S), \tag{95}$$

where $C_{\mathcal{F}}$ and $C_{\mathcal{G}}$ are uniform bounds on the functions in $\mathcal{F}$ and $\mathcal{G}$. Since $\mathcal{F}$ consists of indicator functions, we have $C_{\mathcal{F}} = 1$. Since $G_b(x, y) = (2y - b^\top x)b^\top x$, under the boundedness assumptions, we have $|G_b| \le (2C + dC^2)dC^2 = O(d^2C^4)$, so we set $C_{\mathcal{G}} = O(d^2C^4)$.

Next, we bound the Rademacher complexity of each class:

(1) **Complexity of $\mathcal{F}$:** The decision boundary is defined by the quadratic inequality $F_b(u, x) = 2ub^\top x - (b^\top x)^2 > 0$. Using the polynomial embedding trick, we can lift this quadratic function in $d + 1$ dimensions to a linear function in a higher-dimensional space of dimension $D = \frac{(d+2)(d+3)}{2} = O(d^2)$. Since the VC dimension of linear halfspaces in D dimensions is D, we have $d_{\mathrm{VC}}(\mathcal{F}) = O(d^2)$. Applying Sauer's lemma and Massart's lemma, the Rademacher complexity is:

$$\hat{\mathcal{R}}(\mathcal{F}; S) \le O\left(\sqrt{\frac{d_{\mathrm{VC}}(\mathcal{F})\log n}{n}}\right) = O\left(d\sqrt{\frac{\log n}{n}}\right). \tag{96}$$

(2) **Complexity of $\mathcal{G}$:** The functions in $\mathcal{G}$ are of the form $\phi_y(b^\top x) = (2y - b^\top x)b^\top x$. The derivative of $\phi_y(z) = 2yz - z^2$ with respect to $z = b^\top x$ is bounded by $2C + 2dC^2 = O(dC^2)$. Thus, the quadratic mapping is $O(dC^2)$-Lipschitz. By Talagrand's contraction and the complexity of the linear class, we have:

$$\hat{\mathcal{R}}(\mathcal{G}; S) \leq O(dC^2)\hat{\mathcal{R}}(\mathcal{H}; S) = O\left(\frac{d^2C^4}{\sqrt{n}}\right). \tag{97}$$

Substituting these into the product rule gives:

$$\begin{aligned}\hat{\mathcal{R}}(\mathcal{F} \cdot \mathcal{G}; S) &\leq O(d^2C^4) \cdot O\left(d\sqrt{\frac{\log n}{n}}\right) + 1 \cdot O\left(\frac{d^2C^4}{\sqrt{n}}\right)\\ &= O\left(d^3C^4\sqrt{\frac{\log n}{n}}\right).\end{aligned} \tag{98}$$

Applying the Rademacher generalization bound, with probability at least $1 - \delta/2$, we get:

$$\sup_b |\mathbb{E}[h_b] - \hat{\mathbb{E}}[h_b]| \leq O\left(d^3C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\right). \tag{99}$$

Summing the variance and expectation bounds completes the proof. □

Proof of Theorem 3.8: Advantage Generalization.

Advantage Generalization. Let $\Delta^* = R^*_{\mathrm{V}}(q) - R^*_{\mathrm{S}}(q)$ and $\hat{\Delta}(S) = \min_b \hat{R}_{\mathrm{V}}(b, q) - \min_b \hat{R}_{\mathrm{S}}(b, q)$. We bound the difference using the standard infimum inequality:

$$\left|\min_b f(b) - \min_b g(b)\right| \leq \sup_b |f(b) - g(b)|. \tag{100}$$

Applying this to both the vanilla and strategic risks yields:

$$\begin{aligned}|\Delta^* - \hat{\Delta}(S)| &\leq \left|\min_b R_{\mathrm{V}}(b, q) - \min_b \hat{R}_{\mathrm{V}}(b, q)\right| + \left|\min_b R_{\mathrm{S}}(b, q) - \min_b \hat{R}_{\mathrm{S}}(b, q)\right|\\ &\leq \sup_{b \in \mathcal{W}} |R_{\mathrm{V}}(b, q) - \hat{R}_{\mathrm{V}}(b, q)| + \sup_{b \in \mathcal{W}} |R_{\mathrm{S}}(b, q) - \hat{R}_{\mathrm{S}}(b, q)|.\end{aligned} \tag{101}$$

By taking a union bound over the generalization events in Lemma 3.6 and Lemma 3.7 (allocating failure probability $\delta/2$ to each), they hold simultaneously with probability at least $1 - \delta$. Summing the bounds and absorbing the lower-order vanilla generalization bound $O(d^2C^4/\sqrt{n})$ into the strategic bound $O(d^3C^4\sqrt{\log n/n})$ gives the stated result. □

Proof of Theorem 3.9: Excess Risk Guarantee.

Excess Risk Guarantee. Let $\theta \in \{\text{voluntary}, \text{mandatory}\}$ be the regime selected by the joint selection algorithm, and let b be the slope parameter. The excess risk of the algorithm is:

$$R(\mathcal{A}, S) = R_\theta(b, q) - \min(R^*_V(q), R^*_S(q)). \tag{102}$$

Let $\epsilon(n, \delta) = O\left(d^3C^4\sqrt{\frac{\log n}{n}} + C^2\sqrt{\frac{\log(1/\delta)}{n}}\right)$ be the uniform generalization boundary. By a union bound over the generalization events, the following holds with probability at least $1 - \delta$:

$$|R^*_V(\dots) - \hat{R}^*_V(\dots)| \leq \epsilon, \quad |R^*_S(\dots) - \hat{R}^*_S(\dots)| \leq \epsilon, \quad |\Delta^* - \hat{\Delta}(S)| \leq \epsilon. \tag{103}$$

We analyze two cases based on the empirical strategic advantage $\hat{\Delta}(S)$:

(1) **Case 1:** $|\hat{\Delta}(S)| > \epsilon(n,\delta)$**.** Since $|\Delta^* - \hat{\Delta}(S)| \leq \epsilon(n,\delta)$, the estimation error is too small to flip the sign of the advantage. Consequently, the algorithm correctly identifies the optimal disclosure regime (e.g., if $\hat{\Delta}(S) > \epsilon$, then $\Delta^* > 0$, and the algorithm chooses voluntary disclosure, which is the oracle choice). The excess risk is then bounded solely by the within-regime generalization error of the selected regime:

$$R(\mathcal{A}, S) \leq \epsilon(n,\delta). \tag{104}$$

(2) **Case 2:** $|\hat{\Delta}(S)| \leq \epsilon(n,\delta)$**.** In this case, the empirical advantage is small, and the algorithm may select the suboptimal disclosure regime. However, because $|\hat{\Delta}(S)| \leq \epsilon(n,\delta)$ and $|\Delta^* - \hat{\Delta}(S)| \leq \epsilon(n,\delta)$, the true strategic advantage is bounded by:

$$|\Delta^*| \leq |\Delta^* - \hat{\Delta}(S)| + |\hat{\Delta}(S)| \leq 2\epsilon(n,\delta). \tag{105}$$

Thus, the penalty for selecting the wrong regime (which is exactly $|\Delta^*|$) is at most $2\epsilon(n,\delta)$. The total excess risk is the sum of the within-regime estimation error and this selection penalty:

$$R(\mathcal{A}, S) \leq \epsilon(n,\delta) + 2\epsilon(n,\delta) = 3\epsilon(n,\delta). \tag{106}$$

In both cases, the excess risk is bounded by $3\epsilon(n,\delta) = O(\epsilon(n,\delta))$, which proves the theorem. □

Proof of Lemma 3.10: Population Gradient Formula.

Population Gradient Formula. The population risk is:

$$L(b) = \mathbb{E}_{(U,X,Y)\sim\mathcal{P}}\left[Y^2 - \psi(F_b(U,X))G_b(X,Y)\right]. \tag{107}$$

The term Y^2 does not depend on the receiver parameter b, so its gradient is zero. We compute the gradient of the product term $\psi(F_b(U,X))G_b(X,Y)$ using the chain and product rules.

Let $s(b) = b^\top X$. Using this notation:

$$F_b(U,X) = 2Us(b) - s(b)^2, \quad G_b(X,Y) = 2Ys(b) - s(b)^2. \tag{108}$$

The gradient of $s(b)$ with respect to b is $\nabla_b s(b) = X$. Differentiating F_b and G_b using the chain rule gives:

$$\nabla_b F_b(U,X) = (2U - 2s(b))\nabla_b s(b) = 2(U - b^\top X)X, \tag{109}$$

$$\nabla_b G_b(X,Y) = (2Y - 2s(b))\nabla_b s(b) = 2(Y - b^\top X)X. \tag{110}$$

Now, we apply the product rule to the expectation:

$$\begin{aligned}\nabla L(b) &= -\mathbb{E}\left[\nabla_b\left(\psi(F_b(U,X))G_b(X,Y)\right)\right] \\ &= -\mathbb{E}\left[\psi'(F_b(U,X))\nabla_b F_b(U,X)G_b(X,Y) + \psi(F_b(U,X))\nabla_b G_b(X,Y)\right].\end{aligned} \tag{111}$$

Substituting the gradient terms $\nabla_b F_b$ and $\nabla_b G_b$ yields:

$$\nabla L(b) = -2\mathbb{E}\left[\psi'(F_b(U,X))(U - b^\top X)G_b(X,Y)X + \psi(F_b(U,X))(Y - b^\top X)X\right], \tag{112}$$

which matches (23). □

Proof of Theorem 3.11: Bandit-Feedback Regret.

Bandit-Feedback Regret. Let $\hat{L}_\delta(b) = \mathbb{E}_{v\sim\mathbb{B}^d}[L(b+\delta v)]$ be the smoothed population risk, where $\mathbb{B}^d$ is the solid unit ball in $\mathbb{R}^d$. By Stokes' theorem (or the divergence theorem), the expectation of the one-point gradient estimator $\hat{g}_t = \frac{d}{\delta_t} c_t v_t$ over the sphere surface is:

$$\mathbb{E}_{v_t}\left[\frac{d}{\delta_t}\ell_t(b_t+\delta_t v_t)v_t\right] = \nabla\hat{L}_{\delta_t}(b_t), \tag{113}$$

confirming that $\hat{g}_t$ is an unbiased estimator of the gradient of the smoothed risk.

Because the true risk $L(b)$ is H-smooth, the smoothed risk uniformly approximates $L(b)$ to second order:

$$\sup_{b\in\mathcal{W}} |\hat{L}_{\delta_t}(b) - L(b)| \le \frac{H}{2}\delta_t^2. \tag{114}$$

The expected squared norm of the bandit gradient estimator is bounded by:

$$\mathbb{E}[\|\hat{g}_t\|_2^2] = \mathbb{E}\left[\left\|\frac{d}{\delta_t}c_t v_t\right\|_2^2\right] \le \frac{d^2 c_{\max}^2}{\delta_t^2}, \tag{115}$$

where $c_{\max} = O(d^2C^4)$ is the maximum loss value.

We apply the standard OGD analysis to the sequence of smoothed functions $\hat{L}_{\delta_t}$, which inherit the local λ-strong convexity of L. Using the learning rate $\eta_t = \frac{1}{\lambda t}$:

$$\sum_{t=1}^{T}\mathbb{E}[\hat{L}_{\delta_t}(b_t) - \hat{L}_{\delta_t}(b^*)] \le \sum_{t=1}^{T}\frac{\eta_t}{2}\mathbb{E}[\|\hat{g}_t\|_2^2] \le \frac{d^2c_{\max}^2}{2\lambda}\sum_{t=1}^{T}\frac{1}{\delta_t^2 t}. \tag{116}$$

To express this in terms of the true risk $L(b_t)$, we add and subtract the smoothing approximation error:

$$\sum_{t=1}^{T}\mathbb{E}[L(\hat{b}_t) - L(b^*)] \le \sum_{t=1}^{T}\mathbb{E}[\hat{L}_{\delta_t}(b_t) - \hat{L}_{\delta_t}(b^*)] + H\sum_{t=1}^{T}\delta_t^2. \tag{117}$$

Substituting the exploration radius $\delta_t = \delta_0 t^{-1/4}$ gives:

$$\sum_{t=1}^{T}\mathbb{E}[L(\hat{b}_t) - L(b^*)] \le \frac{d^2c_{\max}^2}{2\lambda\delta_0^2}\sum_{t=1}^{T}t^{-1/2} + H\delta_0^2\sum_{t=1}^{T}t^{-1/2}. \tag{118}$$

Approximating the sum $\sum_{t=1}^{T} t^{-1/2} \le 2\sqrt{T}$ gives:

$$\sum_{t=1}^{T}\mathbb{E}[L(\hat{b}_t) - L(b^*)] \le \left(\frac{d^2c_{\max}^2}{\lambda\delta_0^2} + 2H\delta_0^2\right)\sqrt{T}. \tag{119}$$

By setting the exploration parameter $\delta_0 = \sqrt{\frac{dc_{\max}}{\sqrt{2\lambda H}}} = O(\sqrt{d})$, the leading term becomes:

$$\sum_{t=1}^{T}\mathbb{E}[L(\hat{b}_t) - L(b^*)] = O\left(d\sqrt{T}\right). \tag{120}$$

Dividing by T gives the average strategic regret bound $O\left(\frac{d}{\sqrt{T}}\right)$. □

7.2 Proofs for Classification (Section 4)

The following proofs establish the corresponding results for the discrete multiclass classification setting.

Proof of Lemma 4.1: Sender Best Response in Classification.

Sender Best Response in Classification. Let the committed receiver policy be $H = (h, d)$ with default prediction class i (meaning $\hat{y}_t(\emptyset) = i$) and features predictor $h_i(X)$. Senders possess a discrete preferred class $U \in \{1, \ldots, K\}$ and seek to minimize their 0-1 prediction loss $\mathbb{E}[\mathbf{1}\{\hat{Y} \neq U\}]$. We analyze the sender's best response by cases on the preference type U:

(1) **Case 1: $U = i$.** If the sender chooses to withhold features ($D = 0$), the receiver receives an empty message and outputs the default class i. The sender's loss is:

$$\mathbf{1}\{i \neq U\} = \mathbf{1}\{i \neq i\} = 0. \tag{121}$$

Since the 0-1 loss is non-negative, this is the global minimum possible loss. Consequently, withholding features ($D = 0$) is always a utility-maximizing action for Senders with preference $U = i$.

(2) **Case 2: $U \neq i$.** If the sender chooses to withhold ($D = 0$), the receiver outputs the default class i, causing the sender to suffer loss:

$$\mathbf{1}\{i \neq U\} = 1. \tag{122}$$

If the sender instead reveals features ($D = 1$), the features are successfully received with probability $1-q$, in which case the receiver applies the Bayes predictor $h_i(X)$. With probability q, erasure occurs, and the receiver falls back to predicting i. The sender's expected loss from revealing is:

$$\mathbb{E}_\epsilon \left[\mathbf{1}\{\hat{Y} \neq U\}\right] = (1-q)\Pr(h_i(X) \neq U \mid X, Y) + q \cdot 1 = 1 - (1-q)\Pr(h_i(X) = U \mid X, Y). \tag{123}$$

Assuming that the predictor $h_i(X)$ has a non-zero probability of matching the sender's preference U (i.e., $\Pr(h_i(X) = U) > 0$), and since $1 - q > 0$, the expected loss from revealing is strictly less than 1. Therefore, revealing features ($D = 1$) is strictly optimal for Senders with preference $U \neq i$.

This confirms that the sender best response partitions the population into those who withhold features ($U = i$) and those who reveal them ($U \neq i$). □

Proof of Lemma 4.2: Strategic Risk for Classification.

Strategic Risk for Classification. By Lemma 4.1, the sender's best-response strategy partitions the population into two subpopulations: $U = i$ (who withhold features) and $U \neq i$ (who reveal features). Using the law of total expectation, we decompose the receiver's strategic classification risk $R_S^i(q)$ as:

$$R_S^i(q) = \Pr(U = i)\mathbb{E}\left[\mathbf{1}\{\hat{Y} \neq Y\} \;\middle|\; U = i\right] + \Pr(U \neq i)\mathbb{E}\left[\mathbf{1}\{\hat{Y} \neq Y\} \;\middle|\; U \neq i\right]. \tag{124}$$

We evaluate the conditional expectation for each subpopulation:

- **Subpopulation $U = i$:** Senders in this group always withhold features. The receiver always predicts the default class i. The conditional loss is:

$$\mathbb{E}\left[\mathbf{1}\{\hat{Y} \neq Y\} \;\middle|\; U = i\right] = \Pr(Y \neq i \mid U = i). \tag{125}$$

- **Subpopulation $U \neq i$:** Senders in this group reveal features. The features are successfully received with probability $1 - q$ (resulting in prediction $h_i(X)$ and conditional error $r_X^{(-i)}$) and are erased with probability q (resulting in prediction i and conditional error $\Pr(Y \neq i \mid U \neq i)$). The conditional expected loss is:

$$\mathbb{E}\left[\mathbf{1}\{\hat{Y} \neq Y\} \;\middle|\; U \neq i\right] = q\Pr(Y \neq i \mid U \neq i) + (1-q)r_X^{(-i)}. \tag{126}$$

Substituting these terms and $\pi_i = \Pr(U = i)$ into the strategic risk formula yields:

$$R_S^i(q) = \pi_i \Pr(Y \neq i \mid U = i) + (1 - \pi_i) \left[q \Pr(Y \neq i \mid U \neq i) + (1 - q) r_X^{(-i)} \right]. \quad (127)$$

To obtain the second formulation, we utilize the law of total probability for $\Pr(Y \neq i)$:

$$\Pr(Y \neq i) = \pi_i \Pr(Y \neq i \mid U = i) + (1 - \pi_i) \Pr(Y \neq i \mid U \neq i). \quad (128)$$

Subtracting and adding $(1 - \pi_i)(1 - q) \Pr(Y \neq i \mid U \neq i)$ to (127) allows us to group terms:

$$\begin{aligned} R_S^i(q) &= \pi_i \Pr(Y \neq i \mid U = i) + (1 - \pi_i) \Pr(Y \neq i \mid U \neq i) - (1 - \pi_i)(1 - q) \Pr(Y \neq i \mid U \neq i) \\ &\quad + (1 - \pi_i)(1 - q) r_X^{(-i)} \\ &= \Pr(Y \neq i) + (1 - \pi_i)(1 - q) \left(r_X^{(-i)} - \Pr(Y \neq i \mid U \neq i) \right). \end{aligned} \quad (129)$$

This matches the stated formula (30). □

Proof of Theorem 4.3: Vanilla Optimality.

Vanilla Optimality. Let $c^* \in \arg\min_i \Pr(Y \neq i)$ be the vanilla optimal default class. Fix any competitor default class $j \neq c^*$, and define the strategic risk difference function:

$$f_j(q) = R_S^j(q) - R_S^{c^*}(q). \quad (130)$$

By Lemma 4.2, both $R_S^j(q)$ and $R_S^{c^*}(q)$ are affine functions of q (since q appears linearly). Because the difference of two affine functions is itself affine, $f_j(q)$ is an affine function of q over the domain $q \in [0, 1]$.

An affine function on a closed interval $[0, 1]$ must attain its minimum value at one of the two endpoints, i.e.:

$$\min_{q \in [0,1]} f_j(q) = \min\{f_j(0),\ f_j(1)\}. \quad (131)$$

By assumption, the default class c^* is optimal at both $q = 0$ and $q = 1$, which implies:

$$f_j(0) = R_S^j(0) - R_S^{c^*}(0) \geq 0, \quad (132)$$

$$f_j(1) = R_S^j(1) - R_S^{c^*}(1) \geq 0. \quad (133)$$

Since both endpoint values are non-negative, the minimum of the affine function is non-negative:

$$f_j(q) \geq 0 \quad \forall q \in [0, 1]. \quad (134)$$

This implies $R_S^{c^*}(q) \leq R_S^j(q)$ for all competitor default classes $j \in \{1, \ldots, K\}$ and all $q \in [0, 1]$, confirming that c^* is optimal over the entire interval. □

Proof of Theorem 4.4: Piecewise-Linear Noise Partition.

Piecewise-Linear Noise Partition. The optimal strategic risk is defined as the pointwise minimum of the strategic risks over the set of default classes:

$$R_S^*(q) = \min_{i \in \{1,\ldots,K\}} R_S^i(q). \quad (135)$$

By Lemma 4.2, each $R_S^i(q)$ is an affine function of q, representing a line over $[0, 1]$. Therefore, the optimal strategic risk function $R_S^*(q)$ is the pointwise minimum (lower envelope) of a finite collection of K lines.

By standard convex analysis, the pointwise minimum of a finite number of concave (or affine) functions is concave and piecewise-linear. Since there are at most K distinct lines in the collection, the lower envelope can contain at most K distinct linear segments. The intersection points of these segments define at most $K - 1$ transition points in the interval $(0, 1)$. These transition points partition the unit interval $[0, 1]$ into at most K subintervals, on each of which a single default class achieves the minimum risk. □

Proof of Lemma 4.5: Effective Sample Size Concentration.

Effective Sample Size Concentration. Fix a default class $i \in \{1, \ldots, K\}$. The indicator variables $I_j = \mathbf{1}\{u_j \neq i\}$ are independent and identically distributed Bernoulli random variables with success probability:

$$\mathbb{E}[I_j] = \Pr(U \neq i) \geq \gamma, \tag{136}$$

where we used the nondegenerate mass Assumption 1. The revealed subpopulation sample size is $n_i = \sum_{j=1}^{n} I_j$, which is a Binomial random variable with expectation $\mathbb{E}[n_i] \geq \gamma n$.

We analyze the probability of a large negative deviation of the empirical sample proportion n_i/n from its expectation. Applying Hoeffding's inequality yields:

$$\Pr\left(\frac{n_i}{n} - \mathbb{E}\left[\frac{n_i}{n}\right] \leq -\epsilon\right) \leq \exp\left(-2n\epsilon^2\right). \tag{137}$$

Setting $\epsilon = \gamma/2$ gives:

$$\Pr\left(\frac{n_i}{n} - \Pr(U \neq i) \leq -\frac{\gamma}{2}\right) \leq \exp\left(-\frac{n\gamma^2}{2}\right). \tag{138}$$

Since $\Pr(U \neq i) \geq \gamma$, the event $\frac{n_i}{n} \leq \frac{\gamma}{2}$ implies $\frac{n_i}{n} - \Pr(U \neq i) \leq -\frac{\gamma}{2}$. Thus, we have:

$$\Pr\left(n_i < \frac{\gamma n}{2}\right) \leq \exp\left(-\frac{n\gamma^2}{2}\right). \tag{139}$$

To ensure that this holds for all default classes simultaneously, we take a union bound over the K default classes:

$$\Pr\left(\exists i \in \{1, \ldots, K\} : n_i < \frac{\gamma n}{2}\right) \leq K \exp\left(-\frac{n\gamma^2}{2}\right). \tag{140}$$

For a sufficiently large sample size $n \geq \frac{2}{\gamma^2} \log\left(\frac{2K}{\delta}\right)$, the right-hand side is at most $\delta/2$. Consequently, with probability at least $1 - \delta/2$, every default class receives at least $\frac{\gamma n}{2}$ training samples. □

Proof of Lemma 4.6: Natarajan's Sauer-Shelah Lemma.

Natarajan's Sauer-Shelah Lemma. Let $\mathcal{H}$ be a hypothesis class of functions from $\mathcal{X}$ to $\mathcal{Y} = \{1, \ldots, K\}$ with Natarajan dimension d_N. We prove the growth function bound by induction on the sample size n and the Natarajan dimension d_N.

For the base cases:

- If $n = 0$, the sample is empty, and $|\mathcal{H}|_S| = 1$. The bound holds since $\Phi_K(d_N, 0) = 1$.
- If $d_N = 0$, the class can shatter no points, meaning it can only produce a single labeling, so $|\mathcal{H}|_S| = 1$. The bound holds since $\Phi_K(0, n) = 1$.

Assume the bound holds for all samples of size $n - 1$. Let $S = \{x_1, \ldots, x_n\}$ be a sample of size n, and let $S' = S \setminus \{x_n\}$. We partition the set of realizable labelings $\mathcal{H}|_S$ by their restrictions to S'. For each prefix labeling $y \in \mathcal{H}|_{S'}$, let $C(y) \subseteq \{1, \ldots, K\}$ be the set of possible labels for the final point x_n that are consistent with y. The total number of labelings is:

$$|\mathcal{H}|_S| = \sum_{y \in \mathcal{H}|_{S'}} |C(y)| = \sum_{y \in \mathcal{H}|_{S'}} 1 + \sum_{y \in \mathcal{H}|_{S'}} (|C(y)| - 1). \tag{141}$$

The first term is exactly the number of labelings on S', which is bounded by $\Phi_K(d_N, n - 1)$ by the inductive hypothesis.

For the second term, we utilize the algebraic identity that for any set C of size $k \geq 1$:

$$k - 1 \leq \sum_{a<b} \mathbf{1}\{a \in C \text{ and } b \in C\}. \tag{142}$$

Applying this to $C(y)$ gives:

$$\sum_{y \in \mathcal{H}|_{S'}} (|C(y)| - 1) \leq \sum_{y \in \mathcal{H}|_{S'}} \sum_{a<b} \mathbf{1}\{a, b \in C(y)\} = \sum_{a<b} \sum_{y \in \mathcal{H}|_{S'}} \mathbf{1}\{a, b \in C(y)\}. \tag{143}$$

For any fixed pair of classes (a, b), the inner sum counts the number of prefixes y that can be extended by both a and b at x_n. Let $\mathcal{H}_{a,b}$ be the subset of prefix labelings with this property. If a subset of points $T \subseteq S'$ is N-shattered by $\mathcal{H}_{a,b}$, then the set $T \cup \{x_n\}$ is N-shattered by $\mathcal{H}|_S$ (using the same shattering patterns on T, and using a and b at x_n). Therefore, the Natarajan dimension of $\mathcal{H}_{a,b}$ is at most $d_N - 1$.

Applying the inductive hypothesis to $\mathcal{H}_{a,b}$ for each of the $\binom{K}{2}$ pairs yields:

$$\sum_{y \in \mathcal{H}|_{S'}} (|C(y)| - 1) \leq \binom{K}{2} \Phi_K(d_N - 1, n - 1). \tag{144}$$

Combining the two terms:

$$|\mathcal{H}|_S| \leq \Phi_K(d_N, n - 1) + \binom{K}{2} \Phi_K(d_N - 1, n - 1). \tag{145}$$

Using Pascal's recurrence relation $\binom{n-1}{i} + \binom{n-1}{i-1} = \binom{n}{i}$, we expand this expression:

$$\begin{aligned}
|\mathcal{H}|_S| &\leq \sum_{i=0}^{d_N} \binom{n-1}{i} \binom{K}{2}^i + \binom{K}{2} \sum_{j=0}^{d_N - 1} \binom{n-1}{j} \binom{K}{2}^j \\
&= \sum_{i=0}^{d_N} \binom{n-1}{i} \binom{K}{2}^i + \sum_{i=1}^{d_N} \binom{n-1}{i-1} \binom{K}{2}^i \\
&= \binom{n-1}{0} \binom{K}{2}^0 + \sum_{i=1}^{d_N} \left(\binom{n-1}{i} + \binom{n-1}{i-1} \right) \binom{K}{2}^i \\
&= \sum_{i=0}^{d_N} \binom{n}{i} \binom{K}{2}^i = \Phi_K(d_N, n).
\end{aligned} \tag{146}$$

This completes the inductive proof. □

Proof of Lemma 4.7: Fixed-Default Uniform Convergence.

Fixed-Default Uniform Convergence. Fix a default class $i \in \{1, \dots, K\}$. The difference between the strategic risk $R_S^i(h, q)$ and the empirical strategic risk $\hat{R}_S^i(h, q)$ is:

$$\begin{aligned}
R_S^i(h, q) - \hat{R}_S^i(h, q) = &\left(\pi_i \Pr(Y \neq i \mid U = i) - \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\{u_j = i\} \mathbf{1}\{i \neq y_j\} \right) \\
&+ (1 - \pi_i) q \Pr(Y \neq i \mid U \neq i) - q \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\{u_j \neq i\} \mathbf{1}\{i \neq y_j\} \\
&+ (1 - \pi_i)(1 - q) R_i(h) - (1 - q) \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\{u_j \neq i\} \mathbf{1}\{h(x_j) \neq y_j\}.
\end{aligned} \tag{147}$$

The first two terms do not depend on the classifier h and concentrate to their expectations at a rate of $O(1/\sqrt{n})$ by Hoeffding's inequality.

To bound the third term uniformly, we focus on the classification error of the revealed subpopulation:

$$|R_i(h) - \hat{R}_i(h)| = \left| \Pr(h(X) \neq Y \mid U \neq i) - \frac{1}{n_i} \sum_{j:u_j \neq i} \mathbf{1}\{h(x_j) \neq y_j\} \right|, \tag{148}$$

where n_i is the number of revealed samples.

By the standard Rademacher generalization bound for binary 0-1 loss classes, with probability at least $1 - \delta/K$, we have:

$$\sup_{h \in \mathcal{H}_i} |R_i(h) - \hat{R}_i(h)| \leq 2\hat{\mathcal{R}}_{n_i}(\mathcal{L}_i; S) + 3\sqrt{\frac{\log(2K/\delta)}{2n_i}}, \tag{149}$$

where $\mathcal{L}_i = \{(x, y) \mapsto \mathbf{1}\{h(x) \neq y\} : h \in \mathcal{H}_i\}$.

To bound the empirical Rademacher complexity $\hat{\mathcal{R}}_{n_i}(\mathcal{L}_i; S)$, we project the loss class onto the n_i revealed points. Since the 0-1 loss function is binary-valued, the empirical loss vectors lie in $\{0, 1\}^{n_i}$. Applying Massart's lemma yields:

$$\hat{\mathcal{R}}_{n_i}(\mathcal{L}_i; S) \leq \sqrt{\frac{2 \log |\mathcal{L}_i|_{n_i}|}{n_i}}. \tag{150}$$

The number of distinct loss vectors $|\mathcal{L}_i|_{n_i}|$ is bounded by the number of distinct labelings the hypothesis class $\mathcal{H}_i$ can produce on the n_i points. A multiclass linear predictor with K classes and d features per class has Natarajan dimension $d_N = O(Kd)$. Using Lemma 4.6 (Natarajan's lemma):

$$|\mathcal{H}_i|_{n_i}| \leq \sum_{j=0}^{d_N} \binom{n_i}{j} \binom{K}{2}^j \leq \left(\frac{e n_i K^2}{2 d_N} \right)^{d_N}. \tag{151}$$

Taking the logarithm yields $\log |\mathcal{L}_i|_{n_i}| \leq d_N \log \left(\frac{e n_i K^2}{2 d_N} \right) = O(Kd \log n_i)$. Substituting this back into Massart's bound gives:

$$\hat{\mathcal{R}}_{n_i}(\mathcal{L}_i; S) \leq O\left(\sqrt{\frac{Kd \log n_i}{n_i}} \right). \tag{152}$$

By Lemma 4.5, the sample size concentrates such that $n_i \geq \frac{\gamma n}{2}$ with high probability. Substituting this lower bound into the generalization inequality yields:

$$\sup_{h \in \mathcal{H}_i} |R_S^i(h, q) - \hat{R}_S^i(h, q)| \leq O\left(\sqrt{\frac{Kd \log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}} \right), \tag{153}$$

completing the proof. □

Proof of Lemma 4.8 (Uniform Convergence Over All Defaults). Let E_i be the event that the uniform convergence bound in Lemma 4.7 holds for default class i. By Lemma 4.7, we have:

$$\Pr(E_i^c) \leq \frac{\delta}{K}. \tag{154}$$

Taking a union bound over the K possible default classes:

$$\Pr\left(\exists i \in \{1, \ldots, K\} : E_i^c\right) \leq \sum_{i=1}^{K} \Pr(E_i^c) \leq K \cdot \frac{\delta}{K} = \delta. \tag{155}$$

Therefore, with probability at least $1-\delta$, the event $\bigcap_{i=1}^{K} E_i$ holds, meaning the uniform generalization bound holds simultaneously for all default classes. □

Proof of Theorem 4.9 (Excess Risk of the Learned Strategy). Let i^* and h^* be the oracle optimal default class and classifier, which minimize the true strategic risk:

$$R_S^*(q) = R_S^{i^*}(h^*, q) = \min_i \min_{h \in \mathcal{H}_i} R_S^i(h, q). \tag{156}$$

The empirical learner selects the pair $(\hat{i}^*, \hat{h})$ that minimizes the empirical strategic risk, meaning:

$$\hat{R}_S^{\hat{i}^*}(\hat{h}, q) \le \hat{R}_S^i(h, q) \quad \forall i, h. \tag{157}$$

In particular, this holds for the oracle optimal pair:

$$\hat{R}_S^{\hat{i}^*}(\hat{h}, q) \le \hat{R}_S^{i^*}(h^*, q). \tag{158}$$

By Lemma 4.8, with probability at least $1 - \delta$, the generalization error is uniformly bounded by:

$$\epsilon_n = O\left(\sqrt{\frac{Kd \log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \tag{159}$$

for all defaults and classifiers simultaneously.

We decompose the excess risk of the learned strategy as:

$$\begin{aligned} R_S^{\hat{i}^*}(\hat{h}, q) - R_S^*(q) = &\left(R_S^{\hat{i}^*}(\hat{h}, q) - \hat{R}_S^{\hat{i}^*}(\hat{h}, q)\right) + \left(\hat{R}_S^{\hat{i}^*}(\hat{h}, q) - \hat{R}_S^{i^*}(h^*, q)\right) \\ &+ \left(\hat{R}_S^{i^*}(h^*, q) - R_S^{i^*}(h^*, q)\right). \end{aligned} \tag{160}$$

We bound each of the three terms:

- The first term is bounded by ϵ_n due to uniform generalization of the selected pair.
- The second term is non-positive (≤ 0) due to the empirical optimality of the chosen pair.
- The third term is bounded by ϵ_n due to generalization of the oracle pair.

Combining these bounds yields:

$$R_S^{\hat{i}^*}(\hat{h}, q) - R_S^*(q) \le \epsilon_n + 0 + \epsilon_n = 2\epsilon_n. \tag{161}$$

Substituting the expression for ϵ_n completes the proof. □

Proof of Lemma 4.10 (EXP3 Regret Bound). Let $L_{t,i} = \tilde{\ell}_t(i, \Theta_{t,i}) \in [0, B]$ be the true strategic surrogate loss for arm i at round t. The EXP3 algorithm maintains weights $w_{t,i}$ and updates them using the importance-weighted loss estimates:

$$\hat{L}_{t,i} = \frac{L_{t,i}}{p_{t,i_t}} \mathbf{1}\{i = i_t\}. \tag{162}$$

Taking the expectation conditioned on the selection probabilities $p_{t,i}$:

$$\mathbb{E}_{i_t}[\hat{L}_{t,i}] = p_{t,i} \frac{L_{t,i}}{p_{t,i}} + \sum_{j \ne i} 0 = L_{t,i}, \tag{163}$$

confirming that the estimates are unbiased.

By the standard potential function analysis of the Exponential Weights algorithm using $W_t = \sum_{i=1}^{K} w_{t,i}$:

$$\log \frac{W_{T+1}}{W_1} \le -\frac{\gamma}{K} \sum_{t=1}^{T} \sum_{i=1}^{K} p_{t,i} \hat{L}_{t,i} + \frac{\gamma^2}{2K^2} \sum_{t=1}^{T} \sum_{i=1}^{K} p_{t,i} \hat{L}_{t,i}^2. \tag{164}$$

We lower bound the partition sum using any fixed comparator arm i^*:

$$\log \frac{W_{T+1}}{W_1} \geq \log \frac{w_{T+1,i^*}}{K} = -\frac{\gamma}{K} \sum_{t=1}^{T} \hat{L}_{t,i^*} - \log K. \tag{165}$$

Combining the bounds and multiplying by $\frac{K}{\gamma}$:

$$\sum_{t=1}^{T} \sum_{i=1}^{K} p_{t,i} \hat{L}_{t,i} - \sum_{t=1}^{T} \hat{L}_{t,i^*} \leq \frac{K \ln K}{\gamma} + \frac{\gamma}{2K} \sum_{t=1}^{T} \sum_{i=1}^{K} p_{t,i} \hat{L}_{t,i}^2. \tag{166}$$

Taking expectations on both sides, the first term on the left is $\mathbb{E}\left[\sum_t \tilde{\ell}_t(i_t, \Theta_{t,i_t})\right]$ and the second term is $\mathbb{E}\left[\sum_t \tilde{\ell}_t(i^*, \Theta_{t,i^*})\right]$. For the squared term on the right:

$$\mathbb{E}[\hat{L}_{t,i}^2] = \mathbb{E}\left[\frac{L_{t,i}^2}{p_{t,i}^2} \mathbf{1}\{i = i_t\}\right] = \mathbb{E}\left[\frac{L_{t,i}^2}{p_{t,i}}\right] \leq B\mathbb{E}\left[\frac{L_{t,i}}{p_{t,i}}\right]. \tag{167}$$

Summing this over i gives:

$$\sum_{i=1}^{K} p_{t,i} \mathbb{E}[\hat{L}_{t,i}^2] \leq B \sum_{i=1}^{K} L_{t,i} \leq BK\tilde{\ell}_t(i_t, \Theta_{t,i_t}) \leq B^2 K. \tag{168}$$

Substituting this back into the expectation inequality gives:

$$\mathbb{E}\left[\sum_{t=1}^{T} \tilde{\ell}_t(i_t, \Theta_{t,i_t})\right] - \mathbb{E}\left[\sum_{t=1}^{T} \tilde{\ell}_t(i^*, \Theta_{t,i^*})\right] \leq B\gamma T + \frac{BK \ln K}{\gamma}, \tag{169}$$

which is the stated bound. □

Proof of Lemma 4.11 (Importance-Weighted OGD Regret). Let i^* be a fixed default class. The predictor Θ_{t,i^*} is updated using the gradient $\hat{g}_t \mathbf{1}\{i_t = i^*\}$ which is non-zero only when $i_t = i^*$. The OGD update is $\Theta_{t+1,i^*} = \Pi_{\mathcal{W}}\left(\Theta_{t,i^*} - \eta \hat{g}_{t,i^*}\right)$, where:

$$\hat{g}_{t,i^*} = \frac{\mathbf{1}\{i_t = i^*\}}{p_{t,i^*}} \nabla \phi(\Theta_{t,i^*}; x_t, y_t). \tag{170}$$

By the standard OGD analysis on Euclidean projections:

$$\|\Theta_{t+1,i^*} - \Theta^*\|_2^2 \leq \|\Theta_{t,i^*} - \Theta^*\|_2^2 - 2\eta \langle \hat{g}_{t,i^*},\ \Theta_{t,i^*} - \Theta^* \rangle + \eta^2 \|\hat{g}_{t,i^*}\|_2^2. \tag{171}$$

Rearranging terms:

$$2\eta \langle \hat{g}_{t,i^*},\ \Theta_{t,i^*} - \Theta^* \rangle \leq \|\Theta_{t,i^*} - \Theta^*\|_2^2 - \|\Theta_{t+1,i^*} - \Theta^*\|_2^2 + \eta^2 \|\hat{g}_{t,i^*}\|_2^2. \tag{172}$$

Taking the expectation conditioned on the history up to round t:

$$\mathbb{E}_{i_t}[\hat{g}_{t,i^*}] = \nabla \tilde{\ell}_t(i^*, \Theta_{t,i^*}). \tag{173}$$

Thus, the expected inner product is:

$$\mathbb{E}_{i_t}[\langle \hat{g}_{t,i^*},\ \Theta_{t,i^*} - \Theta^* \rangle] = \langle \nabla \tilde{\ell}_t(i^*, \Theta_{t,i^*}),\ \Theta_{t,i^*} - \Theta^* \rangle. \tag{174}$$

Since $\tilde{\ell}_t(i^*, \Theta)$ is convex in Θ (as the cross-entropy surrogate ϕ is convex), we have:

$$\tilde{\ell}_t(i^*, \Theta_{t,i^*}) - \tilde{\ell}_t(i^*, \Theta^*) \leq \langle \nabla \tilde{\ell}_t(i^*, \Theta_{t,i^*}),\ \Theta_{t,i^*} - \Theta^* \rangle. \tag{175}$$

Next, we bound the expected squared norm of the stochastic gradient:

$$\mathbb{E}_{i_t}[\|\hat{g}_{t,i^*}\|_2^2] = \mathbb{E}_{i_t}\left[\frac{\mathbf{1}\{i_t = i^*\}}{p_{t,i^*}^2} \|\nabla \phi\|_2^2\right] = \frac{1}{p_{t,i^*}} \|\nabla \phi\|_2^2 \leq \frac{K}{\gamma} G^2, \tag{176}$$

where we used the Lipschitz bound $\|\nabla\phi\|_2 \leq G$ and the EXP3 exploration lower bound $p_{t,i^*} \geq \gamma/K$.
Substituting these back into the expectation inequality and summing over $t = 1, \ldots, T$:

$$2\eta \sum_{t=1}^{T} \mathbb{E}[\tilde{L}_t(i^*, \Theta_{t,i^*}) - \tilde{L}_t(i^*, \Theta^*)] \leq \mathbb{E}[\|\Theta_{1,i^*} - \Theta^*\|_2^2] + \eta^2 \sum_{t=1}^{T} \frac{K}{\gamma} G^2 \leq D^2 + \frac{\eta^2 G^2 KT}{\gamma}. \tag{177}$$

Dividing by 2η gives the OGD regret bound. □

Proof of Theorem 4.12 (Classification Regret Bound). The total expected strategic surrogate regret decomposes into the EXP3 selection regret and OGD optimization regret:

$$\begin{aligned}
\text{Regret} &= \mathbb{E}\left[\sum_{t=1}^{T} \tilde{\ell}_t(i_t, \Theta_{t,i_t})\right] - \sum_{t=1}^{T} \tilde{\ell}_t(i^*, \Theta^*) \\
&= \left(\mathbb{E}\left[\sum_{t=1}^{T} \tilde{\ell}_t(i_t, \Theta_{t,i_t})\right] - \mathbb{E}\left[\sum_{t=1}^{T} \tilde{\ell}_t(i^*, \Theta_{t,i^*})\right]\right) \\
&\quad + \left(\mathbb{E}\left[\sum_{t=1}^{T} \tilde{\ell}_t(i^*, \Theta_{t,i^*})\right] - \sum_{t=1}^{T} \tilde{\ell}_t(i^*, \Theta^*)\right).
\end{aligned} \tag{178}$$

Using Lemma 4.10 and Lemma 4.11, the total regret is bounded by:

$$\text{Regret} \leq B\gamma T + \frac{BK \ln K}{\gamma} + \frac{D^2}{2\eta} + \frac{\eta G^2 KT}{2\gamma}. \tag{179}$$

Substituting the learning rate $\eta = \frac{D}{G}\sqrt{\frac{\gamma}{KT}}$:

$$\text{Regret} \leq B\gamma T + \frac{BK \ln K}{\gamma} + DG\sqrt{\frac{KT}{\gamma}}. \tag{180}$$

By setting the exploration rate $\gamma = \min\left\{1, \left(\frac{K \ln K}{T}\right)^{1/3}\right\}$, the regret terms balance:

$$B\gamma T = BT^{2/3}(K \ln K)^{1/3}, \tag{181}$$

$$\frac{BK \ln K}{\gamma} = BT^{2/3}(K \ln K)^{1/3}, \tag{182}$$

and the OGD term becomes $DGT^{2/3}K^{1/3}(\ln K)^{-1/6}$. Summing these terms yields:

$$\text{Regret} \leq O\left(BT^{2/3}(K \ln K)^{1/3} + DGT^{2/3}K^{1/3}\right), \tag{183}$$

which is the sublinear regret rate of $\tilde{O}(T^{2/3})$. □

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